A Noisy Volunteer's Dilemma

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Abstract

The Volunteer's Dilemma asks when individuals will incur a private cost to report a common problem. I alter this setting by adding reporting noise. Strikingly, when reporting costs are bounded away from zero, an arbitrarily small amount of noise prevents any information aggregation in large groups. Further, full aggregation requires the cost distribution to be nonsmooth at zero: for instance, a positive fraction of individuals whose reporting cost is exactly zero. Finally, reporting noise ensures that information-aggregation failures extend beyond the Volunteer's Dilemma to more general settings.

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1 Introduction

The bystander effect has been extensively studied in the social sciences. In this paradigm, a large group observes a common problem, and each individual decides whether to speak up to alert a decision-maker. Speaking up is privately costly, creating a diffusion of responsibility: each individual hopes that someone else will speak up and is therefore more likely to remain silent. Research on bystander inaction was motivated by the widely publicized failure to report a violent crime in New York and was later formalized through classic experiments (Darley and Latané, 1968). More recently, the bystander effect has been documented in contexts such as school bullying and sexual harassment reporting (Nickerson et al., 2014) and in organizational misconduct and safety reporting (Bazerman, 2022).

Standard game-theoretic models (cf. Diekmann, 1985) attempt to explain these phenomena through a Volunteer's Dilemma as follows. A problem arises, and each individual can

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report the problem to a decision-maker at a private cost, $\nu_i \geq 0$, or remain silent at no cost. If any individual reports the problem, the decision-maker rationally infers that a problem arose and investigates, giving every member of the group a benefit of H. Proposition 1 establishes the standard result that as the group size grows large, the probability that no individual reports the problem converges to the minimal cost-benefit ratio (i.e., $\inf \operatorname{supp}(\nu_i)/H$), which could be small in the context above.

In contrast to standard models, which assume no reports are received when no problem exists, I assume reporting is noisy: for any state of the world, a report is received from an individual with strictly intermediate probability. I provide several microfoundations for this assumption in the discussion in Subsection 3.1. These include players who make errors consistent with Quantal Response Equilibrium (McKelvey and Palfrey, 1995), henceforth QRE, and organizations that intentionally introduce noise into reporting (Boudreau et al., 2023). This paper examines how noisy reporting impacts the bystander effect, both by altering reporting incentives and by influencing how the decision-maker interprets reports.

The first implication of noise is that in large groups the decision-maker may optimally decline to investigate even if reports are made. Suppose, for contradiction, that one report is sufficient to trigger an investigation. Then, in large and noisy groups, a report is received with probability approaching one, implying that an investigation occurs irrespective of the state. However, then the decision-maker does not benefit from reporting. Indeed, organizations often require multiple warnings before acting. Before the two Boeing 737 MAX crashes and the Challenger disaster, the organizations involved received employee reports flagging safety concerns, but no single report carried enough weight to trigger investigations (Leggett and Burridge, 2020; Vaughan, 1996).[1]

Further, in contrast to the case absent noise, individual pivotality goes to zero as the group size grows large, irrespective of the distribution of reporting costs.[2] Consequently, any finite density of reporting costs at zero leaves the decision-maker bounded away from her first-best payoff, and, among threshold rules, her asymptotic payoff exceeds her status quo payoff only when that density is above a positive cutoff (Proposition 3).

To streamline the exposition, the paper presents the results within the simple frame-

[1] Others may interpret failures to investigate as intentional cover-ups rather than as outcomes of a Bayesian mechanism-design problem. In my model, however, the qualitative results still apply to organizations whose risk tolerance differs from that of their employees, provided that there is some confidence level at which the organization would investigate (i.e., the decision-maker may benefit from reporting). In support of my interpretation, even Toyota's well-intentioned andon cord system, which is designed to empower workers to flag problems, effectively requires a second report before management stops the production line, reflecting risk tolerance rather than an intentional cover-up (Edmondson, 2023).

[2] Throughout, I define pivotality as the magnitude of an individual's effect on his expected payoff excluding reporting costs. This is sometimes also referred to as "influence."

work of the Volunteer's Dilemma before turning to generalizations in Section 5. First, the information-aggregation failures extend to settings in which the action and state spaces are arbitrary compact subsets of the reals. Second, these failures continue to hold in the following extensions, which are introduced and discussed in Subsection 5.2:

- reporting costs are state dependent, provided that the fraction of individuals with nonpositive reporting costs is the same across states;
- individuals observe conditionally independent noisy signals rather than the state itself;
- individuals choose not only whether to report but also what to report, provided that the content of each report is also received with noise, as in Blume et al. (2007).

These results have implications for a broad range of collective-action problems. For instance, one could consider settings in which individuals "report" by adding their names to a petition, tweeting about a social movement, or participating in a protest (cf. Battaglini, 2017), and the decision-maker is a politician or a voter.[3] In these settings, individuals participate in the political movement largely because they receive personal pleasure from doing so, since their participation has a vanishingly small impact on outcomes. Similarly, one could consider media reporting. In this setting, some partisan outlets serve as a source of noise, reporting misconduct when none exists or remaining silent when it does. Unbiased outlets act strategically: they weigh the benefit of exposing malfeasance against strictly positive reporting costs, such as government retaliation or accusations of bias. If individuals cannot discern which outlets are partisan, then as the media market grows large, the diffusion of responsibility drives these unbiased outlets to remain silent. Finally, consider community enforcement with noisy monitoring, as in Fudenberg and Wolitzky (2024), where the community must sanction a member who is a defecting type, and individuals bear a cost to report defection, owing to fear of retribution. Because whether the member is a defecting type is a common state inferred from the total number of reports, my model predicts that as the community grows large, the diffusion of responsibility leads individuals not to report, precluding efficient sanctioning.

The rest of the paper is structured as follows. Section 2 provides a motivating example; Section 3 describes the model; Subsection 3.1 defines and provides microfoundations for the noise assumption; Section 4 presents the results in the context of the Volunteer's Dilemma; Section 5 provides generalizations; Section 6 relates these findings to the motivating examples; and Section 7 concludes and discusses policy solutions.

[3]Battaglini (2017) studies information aggregation with dispersed noisy signals and shows that a friction preventing information aggregation is the inability of the decision-maker to commit. In contrast, I show aggregation may fail with commitment and perfect observability as long as reporting itself is noisy.

Related Literature This paper is primarily related to three strands of literature: bystander intervention, information aggregation, and pivotality bounds.

First, this paper contributes to the literature on the Volunteer's Dilemma and bystander intervention. My no-noise benchmark uses the heterogeneous-reporting-cost generalization of the bystander-intervention model of Diekmann (1985) described by Bergstrom (2017) and Battaglini and Palfrey (2025). When this literature considers other forms of heterogeneity, it typically assumes either immoral types who receive no private value from speaking up (Campos-Mercade, 2022) or preference heterogeneity, such as ambiguity aversion (e.g., Guha, 2020), rather than reporting noise.[4]

The most closely related paper is Goeree and Holt (2005), which analyzes a Volunteer's Dilemma under QRE. In a QRE, each individual receives a full-support preference shock favoring one of the two decisions. Given these shocks, the probability that any individual reports is strictly positive, and thus the probability of no report goes to zero for large groups, implying first-best decision-making. However, in their setting a problem always occurs, and thus a report is always needed. When ex ante uncertainty about whether a problem exists is introduced, QRE also implies that reports are generated with positive probability regardless of the state. Therefore, the key question in my model is whether a Bayesian decision-maker is able to infer the state given the reports. My results imply that upon adding this uncertainty to the framework of Goeree and Holt (2005), the decision-maker remains bounded away from the first-best for large groups and, among threshold rules, improves on the payoff obtained when reporting is uninformative only when the density of reporting costs at zero is sufficiently large.

Further, a large experimental literature (see Fischer et al., 2011 for a review) tests the Volunteer's Dilemma and public-goods games more broadly. My results are consistent with the behavioral findings from these studies: when a problem exists, reports are received with high probability in large groups.[5] However, to the best of my knowledge, the informational implications—specifically, whether a decision-maker can use noisy reports to infer the presence of a problem and thereby benefit from reporting—have not been directly tested, as these papers do not consider reporting when a problem does not exist.

Second, this paper connects to the literature on information aggregation in large elections. The Condorcet Jury Theorem demonstrates that majority voting aggregates information perfectly in large groups even though each voter observes only a noisy private signal of the state. Feddersen and Pesendorfer (1996, 1997) show that large elections can aggregate information

[4]Other papers consider uncertainty over the number of players (Hillenbrand and Winter, 2018) or analyze a dynamic version of the game (Bliss and Nalebuff, 1984; Bergstrom, 2017).

[5]Such a result is trivially true in my analysis because the noise implies that a report is received with probability bounded away from zero for each individual, and reports are i.i.d. conditional on ω.

despite strategic voting and heterogeneous preferences. Crucially, voting is costless in both models. In contrast, reporting is costly in my setting, introducing a free-riding friction in which each individual hopes someone else bears the cost. Palfrey and Rosenthal (1983) study a setting with costly voting; however, they preclude aggregate uncertainty about the state of the world, whereas the focus of my analysis is on whether reports are informative about this underlying uncertainty.[6] More recently, several studies have identified specific conditions under which information aggregation can fail in large groups, including common uncertainty about signal distributions (Mandler, 2012), ambiguity aversion (Ellis, 2016), state-dependent election turnout (Ekmekci and Lauermann, 2020), and nonlinear objectives over signal distributions (Barelli et al., 2022). Most closely related are two papers by Martinelli (2006, 2007), which study costly information acquisition before elections. As in Martinelli, declining pivotality in large groups (generalizations of which are discussed below) means that aggregation depends on whether agents can undertake informative actions at arbitrarily low cost. The environments nevertheless differ: Martinelli's voters make information-acquisition decisions while still uncertain about the state and then vote under majority rule, whereas here individuals perfectly observe the common state before deciding whether to incur a report-specific cost, and the payoff bounds apply to arbitrary symmetric report-count rules, general state and action spaces, and message spaces.

Finally, on a methodological level, the analysis builds on the theoretical literature bounding pivotality in large games, such as public-goods or voting games. Under the symmetry and noise conditions considered here, these bounds imply that each individual's pivotality is at most on the order of $1/\sqrt{n}$ (e.g., Al-Najjar and Smorodinsky, 2000; Fudenberg et al., 1998; Becker, 2012). While the combinatorial properties underpinning these limits are central to my analysis, my economic framework differs substantially. Crucially, applications of these bounds in the prior literature (such as the public-goods problem in Al-Najjar and Smorodinsky, 2000) rely on costless direct revelation to map individual pivotality to economic efficiency, an argument that does not alone suffice when reporting itself is costly.[7] A related result in Battaglini and Palfrey (2024) shows that whether collective action succeeds depends on how the required number of volunteers, m_n, compares with $n^{2/3}$; the authors note that this is consistent with the $1/\sqrt{n}$ pivotality bounds.[8] Unlike in these papers, the

[6] Chemmanur and Fedaseyeu (2018) study information aggregation with costly communication and show that large groups fail to aggregate information when communication costs are homogeneous. However, their learning failures rely on a restriction to pure-strategy equilibria, which typically involves a loss of generality given symmetric costs (cf. Diekmann, 1985).

[7] Additionally, unlike standard mechanism-design settings (cf. Holmstrom, 1982) where a principal can use transfers to achieve first-best outcomes despite noise, my setting precludes transfers.

[8] When about m_n individuals volunteer, the number of volunteers fluctuates on the order of $\sqrt{m_n}$, so an individual is pivotal with probability of order $1/\sqrt{m_n}$. The indifference condition implies that the proportion

decision-maker's challenge in my setting is twofold: the mechanism must induce enough individuals to bear the cost of reporting, and the decision-maker must be able to use these reports to adapt to the state.

2 Motivating Example

I begin by describing the problem absent noise and the payoffs the decision-maker obtains in large groups.

A state $\omega \in \{0, 1\}$ is drawn, where $\omega = 1$ represents a problem. Each of $n > 1$ individuals observes ω and can report the problem ($r_i = 1$) or remain silent ($r_i = 0$). Silence incurs no cost, but reporting incurs a cost ν_i, which is private information to individual i; the costs are i.i.d. draws from a distribution $F(\cdot)$ that admits a strictly positive, continuous density on $[\underline{\nu}, \bar{\nu}]$ with $0 < \underline{\nu}$. A decision-maker observes $\sum_i r_i$ and chooses whether to investigate ($a \in \{0, 1\}$). All players receive a benefit $H > \underline{\nu}$ if an investigation occurs when a problem exists and a loss $L < 0$ if an investigation occurs when it does not. Finally, I assume the decision-maker would not investigate at her prior (i.e., $\Pr(\omega = 1)H + \Pr(\omega = 0)L < 0$), and I restrict attention to symmetric equilibria.

Proposition 1 (Benchmark: No Noise)**.** *Fix a distribution of ν_i as described above and assume $a = 1$ if and only if $\sum_i r_i > 0$. In the unique symmetric equilibrium, there is never an investigation when $\omega = 0$, and the probability of no investigation when $\omega = 1$ converges to $\underline{\nu}/H < 1$ as $n \to \infty$.*

Without noise, the decision rule $a = 1 \iff \sum_i r_i > 0$ is implementable without commitment and is the decision rule typically studied in the literature. Further, this exact proof also applies in the case $\underline{\nu} = 0$, implying that there is no inefficiency asymptotically if $\underline{\nu} = 0$. Conditional on $\omega = 1$, the proof below follows directly from standard arguments (Battaglini and Palfrey, 2025); when $\omega = 0$, silence is a dominant action because reporting is costly and an investigation is harmful.

Proof of Proposition 1. When $\omega = 0$, $r_i = 0$ is a dominant action. Now focus on $\omega = 1$. Because reporting becomes less attractive as ν_i increases, there exists a threshold, $\nu^*(n)$, for choosing $r_i = 1$. At $\nu^*(n)$, the individual is indifferent between $r_i = 0$ and $r_i = 1$, as stated

willing to volunteer is of the same order, and equating it with the required proportion m_n/n gives $m_n \asymp n^{2/3}$. In the current paper, the chosen threshold must simultaneously induce costly reporting and make the resulting report-count distributions sufficiently different across states for the decision to respond to the state.

below:

$$\nu^*(n) = H\big(1 - F(\nu^*(n))\big)^{n-1}. \tag{1}$$

At $\underline{\nu}$, the left-hand side is strictly less than the right-hand side, and at $\bar{\nu}$, the right-hand side is strictly less than the left-hand side. Further, on $(\underline{\nu}, \bar{\nu})$, the left-hand side is strictly increasing, and the right-hand side is strictly decreasing, resulting in a unique cutoff within the support of ν_i. Because the right-hand side decreases with n, $\nu^*(n)$ is decreasing in n. Further, as $\nu^*(n)$ is bounded below by $\underline{\nu}$, $\lim_{n\to\infty} \nu^*(n)$ exists. Suppose, toward a contradiction, that $\lim_{n\to\infty} \nu^*(n) > \underline{\nu}$. Then the right-hand side of Equation (1) would converge to zero, a contradiction. Therefore, $\lim_{n\to\infty} \nu^*(n) = \underline{\nu}$. Finally,

$$\Pr(\sum_i r_i = 0 \mid \omega = 1) = (1 - F(\nu^*(n)))^n = \Big(\frac{\nu^*(n)}{H}\Big)^{\frac{n}{n-1}}.$$

Taking limits yields the claimed result. □

I introduce noise as follows. Each individual is independently mechanical with probability ϵ and rational with probability $1 - \epsilon$. A mechanical individual chooses $r_i \in \{0, 1\}$ uniformly at random, independently of all other uncertainty, whereas a rational individual maximizes expected utility. Further, the existence of mechanical individuals is common knowledge. Appendix B provides an alternative interpretation: with probability $\epsilon/2$, an individual has preferences misaligned with those of the decision-maker and always wants investigations, and, with probability $\epsilon/2$, an individual never wants investigations.

Proposition 2 (Benchmark: With Noise)**.** *Fix any $\epsilon > 0$. There exists n^* such that for all $n \geq n^*$ and any threshold rule under which $a = 1 \iff \sum_i r_i \geq T_n$, the decision-maker's payoff is bounded above by the payoff she would obtain absent reporting in any equilibrium.*

The main analysis shows that this learning failure continues to hold for more general utility functions and arbitrary symmetric decision rules for the decision-maker. Further, the analysis provides conditions on the reporting costs under which the decision-maker can benefit from reporting in large groups, unlike in the benchmark case.

Proof of Proposition 2. Each mechanical individual reports $r_i = 1$ with probability $1/2$ regardless of ω, so in any symmetric equilibrium $\Pr(r_i = 1 \mid \omega) \in [\epsilon/2, 1 - \epsilon/2]$ for all ω. An individual is pivotal if and only if $\sum_{j\neq i} r_j = T_n - 1$. As $\sum_{j\neq i} r_j \sim \text{Bin}(n - 1, p_{n,\omega})$ for some

$p_{n,\omega} \in [\epsilon/2, 1-\epsilon/2]$,

$$\Pr\Big(\sum_{j\neq i} r_j = T_n - 1\Big) \leq \frac{C_{\epsilon/2}}{\sqrt{n-1}}, \tag{2}$$

where $C_{\epsilon/2}$ is a constant depending only on ϵ. This constant is derived from the Berry–Esseen Theorem, as formalized in Lemma C3. Further, the benefit of a pivotal report is at most H, so the expected benefit of reporting is bounded above by $HC_{\epsilon/2}/\sqrt{n-1}$. Therefore, there exists an n^* such that $HC_{\epsilon/2}/\sqrt{n-1} < \underline{\nu}$ for every $n \geq n^*$, implying no rational individual reports. As only mechanical types report, their reports are uninformative about ω, implying the decision-maker's payoff is weakly below the payoff she would obtain absent reporting. □

3 Model

There is a collective-action problem with $n > 1$ individuals, $i \in \{1, \ldots, n\}$, an unknown state of the world $\omega \in \Omega \subseteq \mathbb{R}$ drawn from a common prior, and a decision-maker who chooses an action $a \in A \subseteq \mathbb{R}$. The timing of the game is as follows. The decision-maker commits to a decision rule that maps the group's reports to a distribution over actions $\Delta(A)$. Next, all individuals observe ω and the decision rule and simultaneously decide whether to report $(r_i = 1)$ or not $(r_i = 0)$.[9]

The decision-maker's utility given $a \in A$ and $\omega \in \Omega$ is denoted by $u^p(a, \omega)$. I assume that A, Ω are compact and u^p is continuous on $A \times \Omega$. Each individual has a private type $\nu_i \in \mathbb{R}$ drawn i.i.d. according to $F(\cdot)$. Further, each individual chooses $r_i \in \{0, 1\}$ to maximize

$$u^i(a, \omega, \nu_i, r_i) := u^p(a, \omega) - r_i \cdot \nu_i, \tag{3}$$

given the chosen decision rule. The only restriction imposed on the decision rule is a standard symmetry restriction in the Volunteer's Dilemma and voting literatures: the rule can depend on the total number of reports, but not on who reports. Formally, $d : \{0, 1, \ldots, n\} \to \Delta(A)$. I note that the decision rule need not be monotone and need not be ex post efficient.

Throughout, I utilize the following definitions.

[9] The restriction to binary reporting captures many of the motivating examples and is also the standard assumption in the Volunteer's Dilemma literature. Further, Subsection 5.2 discusses a generalization where individuals can encode cheap-talk messages in their reports and shows that the payoff upper bounds continue to hold with appropriately defined noise.

Definition 1. *The decision-maker's first-best payoff,* Π^{FB}*, and status quo payoff,* Π^{SQ}*, are*

$$\Pi^{FB} := \mathbb{E}_{\omega}(\max_{a \in A} u^p(a, \omega)) \qquad \textit{and} \qquad \Pi^{SQ} := \max_{a \in A} \mathbb{E}_{\omega}(u^p(a, \omega)). \tag{4}$$

Therefore, her first-best payoff is the payoff she obtains when she can observe ω, and the status quo payoff is what she obtains absent information from the individuals. To avoid a trivial analysis, I assume $\Pi^{FB} > \Pi^{SQ}$.

Finally, to accommodate equilibrium multiplicity, I evaluate the supremum of the decision-maker's payoff over symmetric equilibria.[10] Denote by $D(n)$ the space of all symmetric decision rules with n individuals, with prototypical element d, and let $D^T(n)$ denote the space of all threshold decision rules with n individuals.[11] I denote the set of all symmetric Bayesian Nash equilibria given decision rule d by $\mathcal{E}(d)$ and define

$$\Pi(n) := \sup_{d \in D(n),\, e \in \mathcal{E}(d)} \mathbb{E}_{e,d}(u^p(a, \omega)) \quad \text{and} \quad \Pi^T(n) := \sup_{d \in D^T(n),\, e \in \mathcal{E}(d)} \mathbb{E}_{e,d}(u^p(a, \omega)). \tag{5}$$

The focus of the analysis will be on the limit as n grows large.

3.1 Defining Noise

The simplest way to incorporate noise into the model is to assume that the distribution of ν_i, the reporting costs, assigns positive probability to individuals who (i) have sufficiently high reporting costs that they never prefer to report and (ii) have sufficiently negative reporting costs that they always report.

Assumption 1. *Define* $B := \max_{a,\omega} u^p(a, \omega) - \min_{a,\omega} u^p(a, \omega)$*. The distribution of* ν_i *satisfies* $\Pr(\nu_i < -B) > 0$ *and* $\Pr(\nu_i > B) > 0$*.*

This assumption will be imposed for the remainder of the paper and implies that, because B bounds any payoff difference a report can generate, there exist individuals who always report and individuals who never report, for any decision rule. Additionally, this assumption is implied by QRE models in which individuals receive full-support preference shocks (cf.

[10] The restriction to symmetric equilibria is standard in the Volunteer's Dilemma and voting literatures. Further, as is standard in communication games, multiple equilibria, such as "babbling" equilibria, also exist. In such equilibria, reports are independent of the state, the decision-maker's action is independent of the reports, and individuals therefore have no incentive to condition their reports on the state.

[11] $D^T(n)$ encompasses all thresholds, $T_n \in \{0, 1, \ldots, n+1\}$, whereby for $a_1, a_0 \in A$, $d_{T_n}(\sum_i r_i) = a_1$ if $\sum_i r_i \geq T_n$ and a_0 otherwise. Restricting to thresholds may be with loss of generality. For instance, Chwe (2010) shows that in certain environments the only truthful mechanisms (defined as mechanisms in which all individuals truthfully reveal their signals about the state) are nonmonotone. In this setting, Lemma 1 implies no mechanism is truthful. Finally, allowing for nonmonotone mechanisms, which typically require commitment, is conservative, as the paper provides upper bounds for the decision-maker's payoff.

Goeree and Holt, 2005). Rather than modeling both the preference shock for choosing a specific action and the cost of reporting, I interpret ν_i as the net cost of reporting. Given this assumption, the following result holds.

Lemma 1. *Under Assumption 1, there exists an $\epsilon_* > 0$ such that for every decision rule and every equilibrium,* $\Pr(r_i = 1 \mid \omega) \in [\epsilon_*, 1 - \epsilon_*]$.

All subsequent results use Assumption 1 only through Lemma 1: once reports and non-reports both occur with probabilities bounded away from zero in every state, the proofs use no other feature of Assumption 1. Thus, any alternative source of noise that delivers the conclusion of Lemma 1 while leaving the rest of the reporting incentives unchanged leads to the same conclusions. Appendix B discusses how behavioral types (as in the motivating example in Section 2) or noisy communication channels that sometimes flip messages (as discussed in Section 6 and Appendix B) would also imply Lemma 1. Further, upon restricting attention to monotone decision rules, Appendix B shows that adding individuals who always want investigations and individuals who never want investigations would result in similar aggregation failures.

Finally, I note that the standard Volunteer's Dilemma already includes situations in which individuals never report because their costs are too high. These individuals always result in false negatives, and such individuals on their own will not result in information-aggregation failures, as one report remains sufficient to trigger an investigation. The introduction of noise implies both false negatives and false positives (those who report when a problem does not occur); their combination is the key ingredient for the paper's information-aggregation failures.

4 Volunteer's Dilemma

This section presents results on a special case of the model in Section 3: the standard Volunteer's Dilemma (cf. Diekmann, 1985). Here, $\omega \in \{0, 1\}$, $a \in \{0, 1\}$, and the payoff differences, as defined in Section 2, are

$$H := u^p(1, 1) - u^p(0, 1) > 0, \qquad L := u^p(1, 0) - u^p(0, 0) < 0. \tag{6}$$

One can interpret $r_i = 1$ (respectively, 0) as reporting (respectively, remaining silent), $a = 1$ (respectively, 0) as an investigation occurring (respectively, not occurring), and $\omega = 1$ as a state in which an investigation is needed and $\omega = 0$ as a state in which it is not. Finally, as in Proposition 1, assume that $\mathbb{E}_\omega(u^p(1, \omega)) < \mathbb{E}_\omega(u^p(0, \omega))$, implying that at her prior, the decision-maker would not investigate.

This section provides asymptotic results and then a numerical example to illustrate the group sizes at which information aggregation fails in practice.

4.1 Information Aggregation

Assumption 1 implies that $\Pr(r_i = 1 \mid \omega)$ is strictly between zero and one in every state. Without Assumption 1, individual pivotality need not go to zero as the group size grows large, for example, when $\underline{\nu} > 0$ in Equation (1). Under Assumption 1, however, individual pivotality converges to zero, as proven below.

Lemma 2. *There exists a constant $M > 0$, depending only on B and ϵ_*, such that for every $n \geq 2$, symmetric decision rule, symmetric equilibrium, and state ω, (i) if $\nu_i > \frac{M}{\sqrt{n}}$, then $r_i = 0$ and (ii) if $\nu_i < -\frac{M}{\sqrt{n}}$, then $r_i = 1$.*

The intuition for this result when considering threshold rules is that an individual's report changes the decision-maker's action only when the remaining $n-1$ reports land exactly at a threshold, so the expected gain from reporting equals the gain from being pivotal multiplied by the probability of being pivotal. By Assumption 1, the reports of others are i.i.d. with variance bounded away from zero, so by arguments in the spirit of the Central Limit Theorem, the probability of any particular count is at most on the order of $1/\sqrt{n}$. Finally, since the maximal utility gain from being pivotal is bounded by some constant, the expected gain from reporting is at most $M/\sqrt{n}$. The proof establishes this lemma for arbitrary symmetric rules and for the general environment considered in Section 3.

Lemma 2 implies that information about ω can come only from $\nu_i \in \left[-\frac{M}{\sqrt{n}}, \frac{M}{\sqrt{n}}\right]$. Individuals whose types lie outside this interval choose their action based on ν_i alone, rather than on information about ω.

To build intuition for the subsequent results on information transmission, assume that individuals file more reports in higher states. Next, note that the decision-maker faces two interrelated problems. First, the probability that any individual's report contains information about ω shrinks as n grows large. Second, the standard deviation of the noise in $\sum_i r_i$ is of order $\sqrt{n}$ asymptotically. Therefore, asymptotically, the signal-to-noise ratio when the decision-maker compares the two states is governed by

$$\lim_{n\to\infty} \frac{\sum_i \Pr(r_i = 1 \mid \omega = 1) - \sum_i \Pr(r_i = 1 \mid \omega = 0)}{\sqrt{n}}. \tag{7}$$

Thus, intuitively, for the decision-maker to obtain her first-best payoff (respectively, exceed her status quo payoff), the difference in expected report counts between $\omega = 1$ and $\omega = 0$ must grow faster than $\sqrt{n}$ (respectively, be of order $\sqrt{n}$). In contrast, absent noise, the

condition for the decision-maker to obtain her first-best payoff is far more lenient: the report count in state $\omega = 1$ need only exceed that in state $\omega = 0$ by one. Finally, when the distribution of ν_i is continuously differentiable around zero, Lemma 2 implies the following upper bound for the limit in Equation (7)

$$\lim_{n\to\infty} \frac{n(F(\frac{M}{\sqrt{n}}) - F(-\frac{M}{\sqrt{n}}))}{\sqrt{n}} = M \cdot \lim_{\delta\to 0^+} \frac{F(\delta) - F(-\delta)}{\delta} = 2 \cdot M \cdot F'(0). \tag{8}$$

Here, the first equality follows by setting $\delta = M/\sqrt{n}$, and the second equality is the definition of a derivative. The following proposition formalizes these intuitions and provides additional bounds for $\Pi(n)$ in terms of Π^{SQ} and Π^{FB}, as defined in Definition 1.

Proposition 3. *Recall that $F(\cdot)$ denotes the distribution of reporting costs. The following hold:*

1. *If there exists a $\delta > 0$ such that $F(\delta) - F(-\delta) = 0$, then there exists $n^* \in \mathbb{N}$ such that for all $n > n^*$, $\Pi(n) = \Pi^{SQ}$.*

2. *Suppose F is differentiable at zero. If $F'(0) = 0$, then $\lim_{n\to\infty} \Pi(n) = \Pi^{SQ}$.*

3. *Suppose F is differentiable at zero and $F'(0) > 0$. Then $\limsup_{n\to\infty} \Pi(n) < \Pi^{FB}$. If, in addition, F is continuously differentiable on an open neighborhood of zero, then, writing $\pi := \Pr(\omega = 1)$, there exists a constant $\overline{F}_0 \in (0, \infty)$, depending only on π, H, L, and $F(0)$, such that*

$$\liminf_{n\to\infty} \Pi^T(n) > \Pi^{SQ} \quad \iff \quad F'(0) > \overline{F}_0. \tag{9}$$

4. *If $\lim_{\delta\to 0^+} [F(\delta) - F(-\delta)] > 0$, then $\lim_{n\to\infty} \Pi(n) = \Pi^{FB}$.*

This proposition states that the decision-maker's payoff is determined by the prevalence of individuals with zero reporting costs and of individuals with costs near zero. If there are no zero-cost individuals and too few individuals with costs arbitrarily close to zero (i.e., Statements 1 and 2), then she obtains her status quo payoff, which was defined as her maximal payoff absent reporting. If the density of costs at zero is sufficiently large, but there is no positive point mass at zero, then her payoff is bounded away from her first-best payoff but can exceed her status quo payoff (Statement 3). Finally, if there is a positive point mass at zero, then she obtains her first-best payoff (Statement 4).

For Statement 1, if such a δ exists, then by Lemma 2, there exists an n^* above which an individual's action is a function of his private type, ν_i, alone. Therefore, the decision-maker

optimally disregards the reports and obtains her status quo payoff. Statement 2 and the upper bound in Statement 3 follow from formalizations of Equations (7) and (8).

The lower bound in Statement 3 requires the construction of an equilibrium in which the decision-maker's payoff exceeds her status quo payoff in the limit. By Equation (7), this necessitates that the difference in reporting probabilities between states $\omega = 1$ and $\omega = 0$ be proportional to $\frac{1}{\sqrt{n}}$. I show that, under threshold rules, the decision-maker's payoff exceeds her status quo payoff if and only if the density at zero is sufficiently large. Finally, Statement 4 states that if the distribution of reporting costs has a positive point mass at zero, the decision-maker's payoff converges to her first-best payoff. The intuition is that any individual with $\nu_i = 0$ can simply choose $r_i = \omega$. Therefore, the difference in the probabilities of reporting $r_i = 1$ does not converge to zero as n grows large, thus satisfying the condition for $\lim_{n\to\infty} \Pi(n) = \Pi^{FB}$ in Equation (7).

4.2 Example

The preceding results show that information aggregation fails in large and noisy groups whenever reporting costs are bounded away from zero. Figure 1 illustrates this finding numerically and shows that the group sizes needed for these asymptotics to apply are reasonable given the motivating examples. Here, I fix a utility function for the decision-maker and parameterize the amount of noise by assuming that, with probability $\epsilon/2$, an individual has a sufficiently negative ν_i and therefore has a dominant strategy to report; with probability $\epsilon/2$, an individual has a sufficiently large ν_i and therefore has a dominant strategy not to report; and with probability $1-\epsilon$, an individual is as in the standard Volunteer's Dilemma. Because solving for the decision-maker's globally optimal decision rule is computationally intractable for large n, the figure plots $\Pi^T(n)$: her expected payoff under an optimally chosen threshold. Even a small amount of noise (such as $\epsilon = 0.01$) can cause aggregation failures for groups of size 100. Given that the motivating applications involve engineering teams that may number in the thousands or students witnessing misconduct in settings with hundreds of bystanders, these group sizes at which aggregation fails appear well within the range encountered in practice.

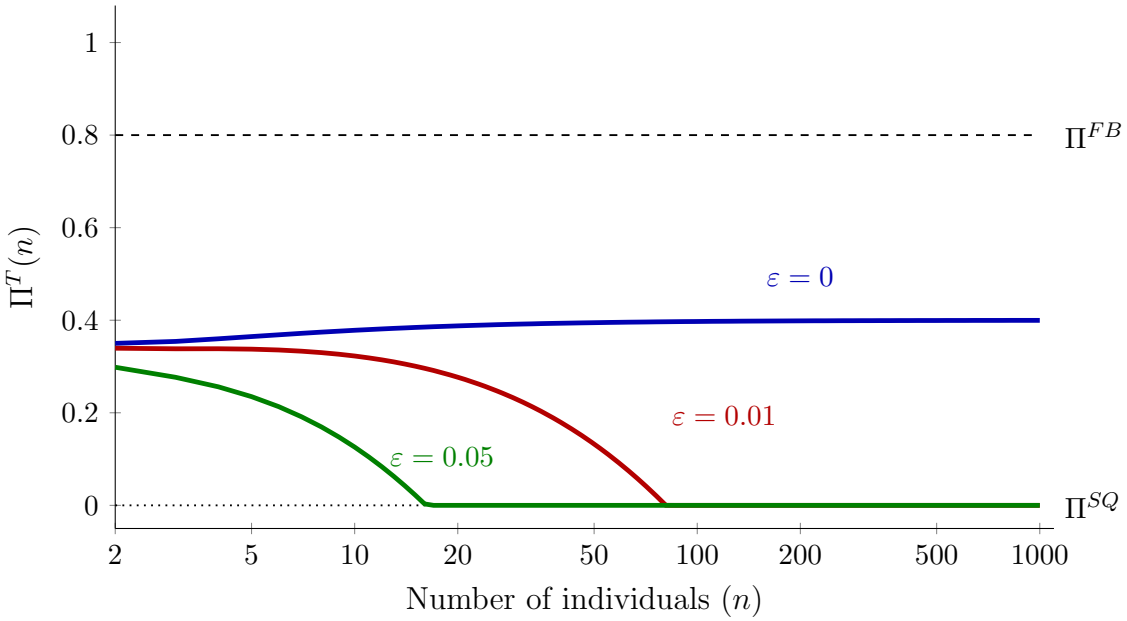


Figure 1: The decision-maker's utility is $u^p(a, \omega) = a(5.5\omega - 1.5)$, where $\omega \in \{0, 1\}$ with $\Pr(\omega = 1) = 0.2$. With probability $\epsilon/2$, ν_i is sufficiently negative that an individual always reports; with probability $\epsilon/2$, ν_i is sufficiently positive that an individual never reports; and with probability $1 - \epsilon$, $\nu_i \sim U[2, 6]$. The blue curve shows the case where $\epsilon = 0$, as in the standard Volunteer's Dilemma, and thus the decision-maker uses a threshold of 1 to choose $a = 1$. The red and green curves show the cases where $\epsilon = 0.01$ and $\epsilon = 0.05$, respectively. In both cases, I plot $\Pi^T(n)$, as defined in Equation (5). Finally, the dashed line at 0.8 corresponds to the first-best payoff, and the dotted line at 0 corresponds to the status quo payoff.

Finally, in the figure, a larger group can improve the decision-maker's payoff when $\epsilon = 0$ but reduce it when $\epsilon > 0$. This is because when $\epsilon = 0$, a larger group size increases the probability that at least one individual has a low-cost realization, which for $n \geq 2$ more than offsets each individual's lower reporting probability. Further, when $\epsilon = 0$, the optimal threshold for large groups remains $T_n = 1$. In contrast, for $\epsilon > 0$, the accumulation of noise in larger groups results in decreased pivotality and an inability of the decision-maker to disentangle true reports from noise, thus decreasing the utility of the decision-maker.

5 Generalizations

This section first expands on the information-aggregation failures described in the previous section. Next, I briefly discuss the results of various extensions in Appendix D: state-dependent reporting costs, worst-case equilibrium selection, noisy observation of the state, and more general message spaces.

5.1 Generalized Model

I begin by showing that the information-aggregation failures persist with the more general preferences, action spaces, and state spaces described in Section 3. Moreover, despite the assumption of binary reports, if there exists a positive probability of individuals with zero reporting cost, the decision-maker achieves her first-best payoff in the limit.

Proposition 4. *Recall that $F(\cdot)$ denotes the distribution of reporting costs. The following hold:*

1. *If there exists a $\delta > 0$ such that $F(\delta) - F(-\delta) = 0$, then there exists a finite cutoff n^* such that for all $n > n^*$, $\Pi(n) = \Pi^{SQ}$.*
2. *If F is differentiable at zero with $F'(0) = 0$, then $\lim_{n\to\infty} \Pi(n) = \Pi^{SQ}$.*
3. *If F is differentiable at zero and $F'(0) > 0$, then $\limsup_{n\to\infty} \Pi(n) < \Pi^{FB}$.*
4. *If $\lim_{\delta\to 0^+} [F(\delta) - F(-\delta)] > 0$, then $\lim_{n\to\infty} \Pi(n) = \Pi^{FB}$.*

This proposition states that the communication breakdowns described in Proposition 3 are not unique to the payoff environment of the Volunteer's Dilemma and occur more generally. The reason is that Lemma 2 (detailing a necessary condition on ν_i for r_i to be informative about ω) does not rely on any details of the environment and, in fact, holds whenever $u^p(a, \omega)$ is bounded. Recall that Lemma 2 implies that reports that convey information about ω can arise only from individuals with types in $\left[-\frac{M}{\sqrt{n}}, \frac{M}{\sqrt{n}}\right]$. The proof of Proposition 4 proceeds by deriving an upper bound on the decision-maker's payoff by ignoring feasibility constraints and assuming these individuals fully convey information about ω. As before, the proportion of such individuals will determine whether information transmission completely fails for finite n (Statement 1), completely fails asymptotically (Statement 2), or is inefficient asymptotically (Statement 3).

Finally, if the distribution of reporting costs has a point mass at zero, the proof of Statement 4 constructs a sequence of decision rules and equilibria under which the decision-maker's payoff converges to the first-best. This is the only construction in the paper that relies on a nonmonotone decision rule. To gain intuition, consider the following strategy profile, which forms an approximate equilibrium for large n: every individual with $\nu_i < 0$ reports, every individual with $\nu_i > 0$ abstains, and every individual with $\nu_i = 0$ independently reports with a carefully chosen state-dependent probability $f(\omega)$. By Lemma 2, any individual's effect on the expected common payoff is on the order of $1/\sqrt{n}$, so the gain from deviating from this profile vanishes as the group grows. The mixing rate $f(\omega)$ can be chosen so that the

aggregate report share allows the decision-maker to infer enough about the state to select a nearly optimal action with probability approaching one. The formal proof constructs such a rule and state-dependent mixing rates while ensuring that the zero-cost types are exactly indifferent in a symmetric equilibrium.

5.2 Extensions

The Online Appendix contains four sets of extensions, which I briefly summarize below.

State-Dependent Reporting Costs It is natural to consider what might occur if the distribution of reporting costs differed depending on the state. In this extension, I consider the binary Volunteer's Dilemma and assume the costs are drawn from F_ω. The key condition for whether the results continue to hold is whether $F_1(0) = F_0(0)$. If these two values are not equal, then the decision-maker always obtains the first-best payoff asymptotically. The intuition is that the fraction of individuals with nonpositive reporting costs (i.e., those who are willing to report even if their report has no effect on decision-making) differs across states, and this difference can be used to tease out the state. If instead $F_1(0) = F_0(0)$, the three upper-bound conclusions of Proposition 3 continue to hold upon replacing the conditions on the behavior of $F(\cdot)$ near zero with conditions on the behavior of $F_1(\cdot), F_0(\cdot)$ near zero. Whether this condition holds is likely context dependent. For instance, one may interpret negative-cost types as a reduced-form representation of individuals biased in favor of investigations (Appendix B.3 provides conditions under which this reduced-form interpretation is valid). Under this interpretation, because the prevalence of these biased types should not depend on the state, $F_1(0) \neq F_0(0)$ can then arise only if the proportion of individuals facing *exactly* zero reporting costs differs across states.

Worst-Case Equilibrium Selection Thus far, the paper has assumed that the decision-maker's preferred equilibrium is selected. One could instead consider a robust approach whereby the decision-maker's worst equilibrium is selected. The upper bounds on the decision-maker's utility continue to hold, shifting the focus to the lower bounds. In the case of the Volunteer's Dilemma, the two lower bounds, qualitatively, continue to hold. The first lower bound in Proposition 3 shows that there exists a threshold on $F'(0)$ above which the decision-maker benefits from reporting asymptotically. An analogous threshold exists in the robust approach. The second lower bound states that if $F(\cdot)$ has a point mass at zero, the decision-maker obtains her first-best payoff asymptotically, and such a result continues to hold in the robust approach. The broad intuition is that if the decision-maker uses a

threshold rule, then reporting is strictly more attractive in state 1 than in state 0 in every equilibrium. A sufficiently high density of near-zero-cost types, or a positive mass at zero, then generates enough separation in report counts to dominate noise uniformly across equilibria.

Noisy Observation of the State The baseline analysis assumes that every individual observes ω perfectly. This extension instead gives each individual an i.i.d. private signal s_i drawn conditional on ω. Under Assumption 1, the three payoff upper bounds continue to hold for any such signal structure. The reason is that the pivotality arguments in the main analysis rest on bounded utility rather than on direct observation of the state, while conditional independence preserves the binomial report-count structure. However, the lower bounds need not hold, as, for instance, if the signals were completely uninformative about ω, then the decision-maker could not improve on her status quo payoff.

General Message Spaces This extension allows for a more general message space. An individual can remain silent at zero cost or, at cost $\nu_i \geq 0$, send any message $m \in [0, 1]$. To maintain the upper-bound results, the analysis requires two types of noise. First, as in Section 2, it requires that mechanical individuals who send pure noise exist with positive probability. Second, it requires noise in how messages are received, as in Blume et al. (2007). This assumption states that if an individual sends message m, the receiver may receive a message $m' \neq m$. Without such an assumption, if the decision-maker ever receives two identical messages, then it is known that these messages come from nonmechanical individuals, completely undoing the noise. Such inference, however, relies on perfect coordination, which this noise removes. With these assumptions, the upper bounds on the decision-maker's payoff continue to hold with more general message spaces.

6 Discussion

The results in the paper show the deleterious effects of reporting noise (i.e., Assumption 1). For instance, in the standard formulation of the Volunteer's Dilemma (where Assumption 1 fails), if the support of ν_i contains zero, then asymptotically the decision-maker obtains her first-best payoff, as can be shown by an argument analogous to that used in Proposition 1. I then show in Subsection 4.1 that this is not the case with Assumption 1. Additionally, absent Assumption 1, the decision-maker's limiting payoff is continuous in $\inf \operatorname{supp}(\nu_i)$ and converges to the first-best payoff as $\inf \operatorname{supp}(\nu_i) \downarrow 0$. In contrast, Subsection 4.1 shows that with Assumption 1, for any differentiable distribution of reporting costs,

the decision-maker's payoff never approaches her first-best payoff and approaches her status quo payoff whenever the distribution of ν_i, $F(\cdot)$, satisfies $F'(0) = 0$. These results imply that noisy reporting in collective-action games leads to significantly less efficient information aggregation.

I now apply these results to the motivating examples (such as reporting harassment, bullying, or safety concerns within organizations). In these contexts, Assumption 1 states both that (i) a decision-maker may observe reports of a problem with positive probability even if no problem exists and that (ii) the decision-maker may observe no reports of a problem with positive probability even if a problem exists. When this assumption holds (for any of the reasons discussed in Subsection 3.1), information aggregation in large groups relies on information from individuals with arbitrarily low reporting costs.

One microfoundation for Assumption 1 is organizations that intentionally add noise to reporting, as in Boudreau et al. (2023). In their setting, adding noise increases plausible deniability for individuals who report on their managers and thus decreases the probability of retaliation from a disgruntled manager. Through the lens of my model, one could view the cost of reporting (i.e., ν_i) as including potential retaliation from a manager. Therefore, while not formally modeled in my analysis, garbling could be viewed as shifting the distribution of ν_i downward, which on its own encourages more reporting. Boudreau et al. (2023) empirically show that doing so can induce reporting, but they do not consider differential effects by team size. Further, such an analysis would be difficult to interpret causally because team size is not exogenous in their context. Consistent with their findings, my analysis suggests that garbling may be helpful for smaller teams by encouraging reports, as it may decrease the cost of reporting. By contrast, I show that in large groups, the same garbling creates a countervailing force: as individual pivotality vanishes, only individuals with reporting costs near zero can transmit information, and the reports received by the decision-maker may fail to be informative about whether a problem occurred.

7 Conclusion

This paper considers a collective-action problem among n individuals who commonly observe a state of the world and then choose whether to submit a report to a decision-maker. The decision-maker then chooses an action based on the beliefs she forms about the state from those reports. I assume that there is noise in the reporting such that reports are made with strictly intermediate probability regardless of the state. With this assumption, even as the number of individuals with arbitrarily low reporting costs grows without bound, I show that the decision-maker's payoff need not exceed her payoff absent any information.

Finally, I show that, in large groups, only individuals with reporting costs arbitrarily close to zero can transmit information.

The mechanism also appears robust to several extensions of the baseline environment. The key force is the pivotality bound in Lemma 2: whenever noise keeps both reporting and nonreporting present in every state, individual influence becomes small in large groups, and the paper's general payoff bounds sharply limit information transmission. This logic suggests that similar failures should occur under (and may be amplified by) nonidentically distributed or correlated shocks (so long as a residual idiosyncratic noise component remains), uncertainty about group size, and preference heterogeneity.

The preceding analysis suggests the following policy implication: organizations facing collective reporting problems subject to noise should either reduce the effective group size or break the symmetry of the decision rule. In contrast, absent noise, the decision-maker's payoff can be increasing in group size (Figure 1 shows the payoff is increasing for all $n > 2$), and Battaglini and Palfrey (2025) show that each individual's utility is weakly increasing in the group size as the equilibrium reporting cutoff falls and reports increasingly derive from lower-cost individuals. With noise, I have shown that larger groups can perform worse than smaller ones and may result in a complete breakdown in information transmission. Both policy levers (reducing group size and introducing asymmetry) can be viewed as concentrating the decision-maker's reliance on a smaller set of identifiable individuals, thereby circumventing the information-aggregation failure. Below, I discuss practical implementations of such policies in specific settings.

The Volunteer's Dilemma applies naturally to settings in which students report bullying in hallways or cafeterias. A common institutional response is to appoint a hallway monitor: a designated individual who is publicly tasked with reporting problems. Through the lens of this model, the decision-maker observes both the sum of the other reports and the monitor's report separately. The monitor, aware of this asymmetry, faces a higher probability of being pivotal and thus has a stronger incentive to report, mitigating the bystander effect.

Analogous practices are widespread in organizational settings. Firms routinely establish dedicated compliance or safety units: small teams that are explicitly charged with identifying and escalating problems and whose reports are treated differently from those of the broader workforce. This effectively reduces the relevant group size to the unit itself and assigns it a privileged role in the decision rule. A related mechanism is the appointment of a devil's advocate: an individual whose designated role is to voice dissent. If the costs of reporting stem in part from deviating from organizational norms, the devil's advocate understands that a report of wrongdoing must originate from him, which sharpens his incentive to report by increasing his perceived pivotality.

A Proofs

Proof of Lemma 1. Let $\epsilon_* := \min\{\Pr(\nu_i < -B), \Pr(\nu_i > B)\} > 0$. Conditional on any state and any reports of the other individuals, changing one's report can change expected utility from the decision-maker's action by at most B in absolute value, since B is the range of u^p. Hence every type with $\nu_i < -B$ strictly prefers to report, and every type with $\nu_i > B$ strictly prefers not to report. Therefore, $\Pr(r_i = 1 \mid \omega) \in [\epsilon_*, 1 - \epsilon_*]$ for every state, decision rule, and equilibrium. □

Proof of Lemma 2. Because types are i.i.d. and all individuals use the same equilibrium strategy, conditional on ω the reports of the other individuals are i.i.d. as well. Fix $n \geq 2$, a symmetric decision rule, and a symmetric equilibrium. Further, due to symmetry, it is sufficient to analyze individual 1. Define $x_{n,\omega} := \sum_{i=2}^{n} r_i \sim \text{Bin}(n-1, p_{n,\omega})$, where $p_{n,\omega} = \Pr(r_i = 1 \mid \omega)$.

Let $U_\omega^k = \mathbb{E}(u^p(a,\omega) \mid \sum_{i=1}^{n} r_i = k, \omega)$ be the individual's gross expected utility when k reports are made, where the expectation is included to allow for a possibly randomized action by the decision-maker. The individual's expected marginal benefit of reporting, $\Delta(\omega)$, is

$$\Delta(\omega) := \sum_{k=1}^{n} U_\omega^k \Pr(x_{n,\omega} = k-1) - \sum_{k=0}^{n-1} U_\omega^k \Pr(x_{n,\omega} = k).$$

Since $\sum_{k=1}^{n} \Pr(x_{n,\omega} = k-1) = \sum_{k=0}^{n-1} \Pr(x_{n,\omega} = k) = 1$, replacing u^p by $u^p + c$ for any constant c shifts every U_ω^k by c and hence leaves $\Delta(\omega)$ unchanged: only differences in u^p affect the reporting incentive. Because $\Delta(\omega)$ is invariant to such shifts, we may center u^p so that $\max_{a,\omega} u^p(a,\omega) = -\min_{a,\omega} u^p(a,\omega) = B/2$, which gives $|U_\omega^k| \leq B/2$ for every k. Summing by parts and applying the triangle inequality,

$$\begin{aligned}
|\Delta(\omega)| &= \Big| U_\omega^n \Pr(x_{n,\omega} = n-1) - U_\omega^0 \Pr(x_{n,\omega} = 0) \\
&\qquad + \sum_{k=1}^{n-1} U_\omega^k \big(\Pr(x_{n,\omega} = k-1) - \Pr(x_{n,\omega} = k) \big) \Big| \\
&\leq \frac{B}{2} \left(\Pr(x_{n,\omega} = n-1) + \Pr(x_{n,\omega} = 0) + \sum_{k=1}^{n-1} \big| \Pr(x_{n,\omega} = k-1) - \Pr(x_{n,\omega} = k) \big| \right).
\end{aligned}$$

Let k^* be a mode of $x_{n,\omega}$. Then $\Pr(x_{n,\omega} = k)$ is nondecreasing in k for $k \leq k^*$ and nonin-

creasing for $k \geq k^*$. Therefore:

$$\begin{aligned}\sum_{k=1}^{n-1} \left| \Pr(x_{n,\omega} = k-1) - \Pr(x_{n,\omega} = k) \right| &= \sum_{k=1}^{k^*} \left(\Pr(x_{n,\omega} = k) - \Pr(x_{n,\omega} = k-1) \right) \\ &\quad + \sum_{k=k^*+1}^{n-1} \left(\Pr(x_{n,\omega} = k-1) - \Pr(x_{n,\omega} = k) \right) \\ &= 2\Pr(x_{n,\omega} = k^*) - \Pr(x_{n,\omega} = 0) - \Pr(x_{n,\omega} = n-1),\end{aligned}$$

as both sums telescope. Substituting this into the previous bound implies

$$|\Delta(\omega)| \leq B \max_k \Pr(x_{n,\omega} = k) \leq \frac{BC}{\sqrt{n-1}} \leq \frac{\sqrt{2}BC}{\sqrt{n}} := \frac{M}{\sqrt{n}},$$

where C is the constant derived in Lemma C3, whose conditions are met given Lemma 1. Type ν_i's net benefit from reporting is $\Delta(\omega) - \nu_i$, so $\nu_i > M/\sqrt{n} \geq |\Delta(\omega)|$ implies $r_i = 0$, and $\nu_i < -M/\sqrt{n}$ implies $r_i = 1$. □

Let Φ and φ denote the standard normal distribution function and density, respectively.

Lemma A1. *Suppose F is continuously differentiable on an open neighborhood of zero, let $t := F'(0) > 0$, set $s := F(0) \in (0,1)$ and $\sigma^2 := s(1-s)$, and write $\pi := \Pr(\omega = 1) \in (0,1)$. Define*

$$\mathcal{U}(t) := \left\{ (u_1, u_0) \in \mathbb{R}^2 : u_1 + \frac{tH}{\sigma^2}\varphi(u_1) = u_0 - \frac{t|L|}{\sigma^2}\varphi(u_0) \right\}, \tag{10}$$

and

$$V(t) := \sup_{(u_1,u_0)\in\mathcal{U}(t)} \left\{ \pi H \Phi(-u_1) - (1-\pi)|L|\Phi(-u_0) \right\}. \tag{11}$$

Then $V(t) \geq 0$ and

$$\lim_{n\to\infty} \Pi^T(n) = \Pi^{SQ} + V(t). \tag{12}$$

Proof of Lemma A1. Set

$$G_1 := H, \quad G_0 := L, \quad \kappa_\omega := \frac{tG_\omega}{\sigma^2}, \quad f_\omega(u) := u + \kappa_\omega \varphi(u),$$

and

$$W(u_1, u_0) := \pi H \Phi(-u_1) - (1-\pi)|L|\Phi(-u_0).$$

Then $\mathcal{U}(t) = \{(u_1, u_0) : f_1(u_1) = f_0(u_0)\}$ and $V(t) = \sup_{\mathcal{U}(t)} W$. Because $f_\omega(u) - u$ is bounded, each f_ω is surjective. For each c, choose $u_\omega(c)$ such that $f_\omega(u_\omega(c)) = c$. As $c \to \infty$, necessarily $u_\omega(c) \to \infty$, so $W(u_1(c), u_0(c)) \to 0$. Hence $V(t) \geq 0$.

Because $a = 0$ is the status quo action, every decision rule and equilibrium satisfies

$$\mathbb{E}_{e,d}(u^p(a, \omega)) = \Pi^{SQ} + \pi H \Pr(a = 1 \mid \omega = 1) - (1-\pi)|L| \Pr(a = 1 \mid \omega = 0),$$

and the last two terms are the gain over Π^{SQ} referred to below.

Consider any sequence of thresholds T_n and symmetric equilibria, with state-ω reporting probability $p_{n,\omega}$. When an investigation takes place if and only if the number of reports is at least T_n, the gross reporting benefit is $G_\omega \rho_{\omega,n}$, where

$$\rho_{\omega,n} := \Pr(\mathrm{Bin}(n-1, p_{n,\omega}) = T_n - 1)\,.$$

Lemma 2 puts this cutoff in the neighborhood where F is continuously differentiable and atomless. Thus, for all large n,

$$p_{n,\omega} = F(G_\omega \rho_{\omega,n}), \qquad p_{n,\omega} = s + O(n^{-1/2}), \tag{13}$$

uniformly over thresholds and equilibria. Define

$$c_n := \frac{T_n - ns}{\sigma\sqrt{n}}, \qquad y_{\omega,n} := \frac{\sqrt{n}(p_{n,\omega} - s)}{\sigma}.$$

The variables $y_{\omega,n}$ are bounded. Along any subsequence, pass to a further subsequence along which c_n converges in the extended real line and both $y_{\omega,n}$ converge. If $c_n \to +\infty$ or $-\infty$, Chebyshev's inequality makes the investigation probability converge, in both states, to zero or one, respectively. The limiting gain over Π^{SQ} is then 0 or $\pi H - (1-\pi)|L| < 0$, hence at most $V(t)$.

Suppose $c_n \to c \in \mathbb{R}$, write $y_{\omega,n} \to y_\omega$, and put $u_\omega := c - y_\omega$. Apply Lemma C3 with $\ell = -1$, $x = \sigma y_{\omega,n}$, and that lemma's t equal to $(T_n - 1 - ns)/\sqrt{n}$, so that $s + x/\sqrt{n} = p_{n,\omega}$, $k_n(t) = T_n - 1$, and $(t - x)/\sigma \to u_\omega$. For the tail, use $\ell = 0$ and $t = (T_n - ns)/\sqrt{n}$. Because the convergence in that lemma is uniform over bounded (x, t), this gives

$$\sqrt{n}\,\rho_{\omega,n} \to \frac{1}{\sigma}\varphi(u_\omega), \qquad \Pr(\mathrm{Bin}(n, p_{n,\omega}) \geq T_n) \to \Phi(-u_\omega).$$

Since $F(x) = s + tx + o(x)$, the fixed-point equation yields $y_\omega = \kappa_\omega \varphi(u_\omega)$, or $f_\omega(u_\omega) = c$. Hence $(u_1, u_0) \in \mathcal{U}(t)$, and the limiting gain is $W(u_1, u_0) \leq V(t)$.

For a rule in which an investigation takes place if and only if the number of reports is less than T_n, the same proof works because the gross reporting benefit is $-G_\omega \rho_{\omega,n}$. If $c_n \to \pm\infty$, the limiting gain is again 0 or $\pi H - (1-\pi)|L|$. If $c_n \to c \in \mathbb{R}$, the same steps give $y_\omega = -\kappa_\omega \varphi(u_\omega)$ and an investigation probability converging to $\Phi(u_\omega)$. Setting $v_\omega := -u_\omega$ gives $f_\omega(v_\omega) = -c$, so $(v_1, v_0) \in \mathcal{U}(t)$, and the limiting gain is $W(v_1, v_0) \leq V(t)$. Finally, for a constant rule, the gain is 0 or $\pi H - (1-\pi)|L| < 0$. Applying these arguments to rules and equilibria whose payoffs are within $1/n$ of $\Pi^T(n)$, along a subsequence attaining the limsup, gives

$$\limsup_{n\to\infty} \Pi^T(n) \leq \Pi^{SQ} + V(t). \tag{14}$$

Call a feasible pair (u_1, u_0) regular if $f_1'(u_1) f_0'(u_0) \neq 0$. We show that regular pairs are dense in $\mathcal{U}(t)$. Feasibility implies $u_0 - u_1 = \kappa_1 \varphi(u_1) - \kappa_0 \varphi(u_0) > 0$. Also, $f_\omega'(u) = 1 - \kappa_\omega u \varphi(u)$. Thus $f_1'(u_1)$ and $f_0'(u_0)$ cannot both be zero: if they were, then $\kappa_1 > 0 > \kappa_0$ would imply $u_1 > 0 > u_0$, contradicting $u_0 > u_1$.

Now consider a nonregular feasible pair. Suppose, without loss of generality, that $f_1'(u_1) = 0$. Then $f_0'(u_0) \neq 0$, so f_0 is strictly monotone in a neighborhood of u_0. Hence, for u_1' sufficiently close to u_1, there is a u_0' close to u_0 such that $f_0(u_0') = f_1(u_1')$. Since $u\varphi(u)$ is unimodal on each half-line, f_1' has only finitely many zeros, so u_1' can be chosen arbitrarily close to u_1 with $f_1'(u_1') \neq 0$. For u_1' sufficiently close to u_1, we also have $f_0'(u_0') \neq 0$. Thus regular feasible pairs can be found arbitrarily close to every nonregular feasible pair, so regular pairs are dense in $\mathcal{U}(t)$.

Fix a regular pair and let $c := f_1(u_1) = f_0(u_0)$. Set

$$T_n := \lceil ns + c\sigma\sqrt{n} \rceil, \quad c_n := \frac{T_n - ns}{\sigma\sqrt{n}} \to c, \quad r_n(u) := s + \frac{\sigma(c_n - u)}{\sqrt{n}}.$$

For large n, $T_n \in \{1, \ldots, n\}$. On a compact neighborhood of u_ω, define

$$\Psi_{\omega,n}(u) := u + \frac{\sqrt{n}}{\sigma} \left[F(G_\omega \Pr(\mathrm{Bin}(n-1, r_n(u)) = T_n - 1)) - s \right].$$

Uniformly on this neighborhood, Lemma C3 and differentiability of F give $\Psi_{\omega,n} \to f_\omega$. Since $f_\omega(u_\omega) = c$ and $f_\omega'(u_\omega) \neq 0$, the Intermediate Value Theorem gives roots $u_{\omega,n} \to u_\omega$ with $\Psi_{\omega,n}(u_{\omega,n}) = c_n$. This is the equilibrium condition for the state-ω reporting probability $r_n(u_{\omega,n})$. Hence the associated cutoff strategies constitute a symmetric equilibrium. Its

payoff converges to $\Pi^{SQ} + W(u_1, u_0)$, so

$$\liminf_{n\to\infty} \Pi^T(n) \geq \Pi^{SQ} + W(u_1, u_0).$$

Taking the supremum over regular pairs gives $\liminf_n \Pi^T(n) \geq \Pi^{SQ} + V(t)$ by density and continuity of W. Together with (14), this proves (12). □

Lemma A2. *Under the assumptions and notation of Lemma A1, there exists $\overline{F}_0 \in (0, \infty)$, depending only on π, H, L, and $F(0)$, such that*

$$V(t) > 0 \quad \Longleftrightarrow \quad t > \overline{F}_0. \tag{15}$$

Proof of Lemma A2. Put $q(u) := \Phi(-u)$, $\ell := |L|$, $\beta_1 := \pi H$, and $\beta_0 := (1-\pi)\ell$. That the decision-maker chooses 0 at her prior implies $\rho := \beta_1/\beta_0 \in (0, 1)$. Define

$$D := \{(u_1, u_0) : \beta_1 q(u_1) > \beta_0 q(u_0)\}.$$

Since q is continuous, D is open. Moreover, if $(u_1, u_0) \in D$, then $q(u_0) < \rho q(u_1) < q(u_1)$. Since q is decreasing, this implies $u_0 > u_1$. Define

$$\tau(u_1, u_0) := \frac{\sigma^2(u_0 - u_1)}{H\varphi(u_1) + \ell\varphi(u_0)}.$$

By the definitions of $V(t)$ and $\mathcal{U}(t)$, $V(t) > 0$ is equivalent to the existence of a point in D satisfying $t = \tau(u_1, u_0)$. Hence

$$S := \{t > 0 : V(t) > 0\} = \tau(D).$$

Further, if $t \in S$, then every $t' > t$ also lies in S. To see this, write $t = \tau(u_1, u_0)$ with $(u_1, u_0) \in D$. Because q is decreasing, $(u_1, x) \in D$ for every $x \geq u_0$. Moreover, $\tau(u_1, x)$ is continuous in x and diverges as $x \to \infty$, so the Intermediate Value Theorem gives an $x \geq u_0$ with $\tau(u_1, x) = t'$.

We next show that $\inf_D \tau > 0$. Fix $(u_1, u_0) \in D$ and put $d := u_0 - u_1$. Since $q(u_0) < \rho q(u_1)$,

$$(1-\rho)q(u_1) < q(u_1) - q(u_0).$$

If $u_1 < 1$, then $q(u_1) \geq q(1)$ and $q(u_1) - q(u_0) \leq \varphi(0)d$. Together with $H\varphi(u_1) + \ell\varphi(u_0) \leq$

$(H+\ell)\varphi(0)$, this gives

$$\tau(u_1,u_0) \geq \frac{\sigma^2(1-\rho)q(1)}{(H+\ell)\varphi(0)^2}.$$

If instead $u_1 \geq 1$, then φ is decreasing on $[u_1,u_0]$, so $q(u_1)-q(u_0) \leq \varphi(u_1)d$ and $H\varphi(u_1)+\ell\varphi(u_0) \leq (H+\ell)\varphi(u_1)$. Further, $q(u_1) \geq u_1^{-1}\varphi(u_1+u_1^{-1}) \geq u_1^{-1}\varphi(u_1)e^{-3/2}$ and $u_1\varphi(u_1) \leq \varphi(0)$, which imply $\varphi(u_1)^2 \leq \varphi(0)e^{3/2}q(u_1)$. Combining these inequalities gives

$$\tau(u_1,u_0) \geq \frac{\sigma^2(1-\rho)}{(H+\ell)\varphi(0)e^{3/2}}.$$

Both lower bounds are positive and independent of (u_1,u_0), so $\inf_D \tau > 0$.

Next, $(0,u_0) \in D$ for all large u_0, so S is nonempty. Finally, $\inf_D \tau$ is not attained. Suppose instead that τ attains its minimum at some $(u_1,u_0) \in D$. Since D is open, both partial derivatives of τ must be zero at this point. Writing $d := u_0 - u_1$ and $P := H\varphi(u_1)+\ell\varphi(u_0)$, these derivatives are

$$\partial_{u_1}\tau = \frac{\sigma^2[-P+dHu_1\varphi(u_1)]}{P^2}, \qquad \partial_{u_0}\tau = \frac{\sigma^2[P+d\ell u_0\varphi(u_0)]}{P^2}.$$

If both were zero, the first equality would imply $u_1 > 0$, while the second would imply $u_0 < 0$. This contradicts $u_0 > u_1$ on D. Hence τ does not attain its infimum on D. Therefore, with $\overline{F}_0 := \inf_D \tau \in (0,\infty)$, we have $S = (\overline{F}_0,\infty)$. Its definition depends only on π, H, L, and $\sigma^2 = F(0)(1-F(0))$. □

Proof of Proposition 4. Throughout, note that the decision-maker can always achieve her status quo payoff by employing a constant decision rule. Recall the definition of B and let $h_n = M/\sqrt{n}$, with M as in Lemma 2. Fix a group size, a symmetric decision rule, and a symmetric equilibrium. Conditional on a state ω, write

$$p_{n,\omega} := \Pr(r_i = 1 \mid \omega), \qquad P_{n,\omega} := \mathrm{Bin}(n, p_{n,\omega}).$$

For notational convenience, the dependence on d and e is suppressed. The gross expected benefit from reporting equals the reporting cutoff. Lemma 2 places that cutoff in $[-h_n, h_n]$. Because a type at the cutoff may mix, monotonicity of F therefore gives, uniformly over states, decision rules, and symmetric equilibria,

$$F((-h_n)^-) \leq p_{n,\omega} \leq F(h_n). \tag{16}$$

Statements 1–3. Let ϵ_* be as in Lemma 1, and define

$$p_n^- := \max\{F((-h_n)^-), \epsilon_*\}, \qquad p_n^+ := \min\{F(h_n), 1-\epsilon_*\}.$$

By (16) and Lemma 1, $p_{n,\omega} \in [p_n^-, p_n^+] \subseteq [\epsilon_*, 1-\epsilon_*]$ uniformly over states, decision rules, and symmetric equilibria. Writing $b_{n,p}(k) := \binom{n}{k} p^k (1-p)^{n-k}$, concavity of

$$p \longmapsto \log b_{n,p}(k) = \log \binom{n}{k} + k \log p + (n-k)\log(1-p)$$

implies, for every $p_{n,\omega} \in [p_n^-, p_n^+]$ and every k,

$$P_{n,\omega}(k) \geq q_n(k) := \min\{b_{n,p_n^-}(k), b_{n,p_n^+}(k)\}.$$

Moreover,

$$\beta_n := \sum_{k=0}^{n} q_n(k) = 1 - \mathrm{TV}\big(\mathrm{Bin}(n, p_n^-), \mathrm{Bin}(n, p_n^+)\big)\,.$$

Let $R_n(k) := q_n(k)/\beta_n$. Then $P_{n,\omega} \geq \beta_n R_n$ pointwise, so for every ω, there is a probability mass function $Q_{n,\omega}$ such that

$$P_{n,\omega} = \beta_n R_n + (1-\beta_n) Q_{n,\omega},$$

where R_n is independent of the state. Hence, applying any decision rule d_n to R_n yields the same action distribution in every state. Denote this distribution by $\mu_n := \sum_{k=0}^{n} R_n(k) d_n(k)$. The decision-maker's payoff is at most

$$\beta_n \int_A \mathbb{E}_\omega[u^p(a,\omega)]\,\mu_n(da) + (1-\beta_n)\mathbb{E}_\omega\left[\max_{a\in A} u^p(a,\omega)\right] \leq \beta_n \Pi^{SQ} + (1-\beta_n)\Pi^{FB}.$$

Since β_n does not depend on the decision rule or the equilibrium, and since the decision-maker can always achieve her status quo payoff,

$$\Pi^{SQ} \leq \Pi(n) \leq \beta_n \Pi^{SQ} + (1-\beta_n)\Pi^{FB}.$$

Each statement now follows from the behavior of β_n.

For Statement 1, choose n^* so that $h_n < \delta$ for every $n > n^*$. The equality $F(\delta) = F(-\delta)$ and monotonicity imply that F is constant on $[-\delta, \delta]$. Hence, for $n > n^*$, both endpoints in (16) equal the same constant, so $p_n^- = p_n^+$ and $\beta_n = 1$. Thus $\Pi(n) = \Pi^{SQ}$ for every $n > n^*$.

For Statements 2 and 3, $F((-h_n)^-) \geq F(-2h_n)$ and differentiability of F at zero imply that

$$\sqrt{n}\,(p_n^+ - p_n^-) \leq \sqrt{n}\big(F(h_n) - F(-2h_n)\big) \longrightarrow 3MF'(0).$$

If $F'(0) = 0$, then $n(p_n^+ - p_n^-)^2 \to 0$, so Lemma C1 implies that $\beta_n \to 1$ and therefore that $\Pi(n) \to \Pi^{SQ}$, which proves Statement 2. If $F'(0) > 0$, then $n(p_n^+ - p_n^-)^2$ is bounded, so Lemma C1 implies that $\beta := \liminf_{n\to\infty} \beta_n > 0$. As $\Pi^{FB} > \Pi^{SQ}$, it follows that

$$\limsup_{n\to\infty} \Pi(n) \leq \beta\Pi^{SQ} + (1-\beta)\Pi^{FB} < \Pi^{FB},$$

which proves Statement 3.

Statement 4.

By right continuity of F, $\alpha := \lim_{\delta\downarrow 0}[F(\delta) - F(-\delta)] = F(0) - F(0^-) > 0$. Write $s_- := F(0^-)$ and $s_+ := F(0) = s_- + \alpha$, so that $s_- < s_+$.

Fix $\xi > 0$. Because A is compact and u^p is uniformly continuous on $A \times \Omega$, there exist actions $a_1, \ldots, a_K \in A$ such that, for every ω,

$$M_K(\omega) := \max_{1\leq j\leq K} u^p(a_j, \omega) \geq \max_{a\in A} u^p(a, \omega) - \xi. \tag{17}$$

Let

$$\bar{d} := \frac{1}{K}\sum_{j=1}^{K} \delta_{a_j}, \;\; \bar{u}(\omega) := \int_A u^p(a,\omega)\,\bar{d}(da) = \frac{1}{K}\sum_{j=1}^{K} u^p(a_j,\omega), \;\; g(\omega) := M_K(\omega) - \bar{u}(\omega) \geq 0.$$

Choose distinct numbers $c_1, \ldots, c_K$ and $\rho > 0$ such that the intervals

$$[c_j - 4\rho, c_j + 4\rho], \qquad j = 1, \ldots, K,$$

are pairwise disjoint and contained in (s_-, s_+). For each n, define the symmetric decision rule

$$d_n(k) := \begin{cases} \delta_{a_j}, & \text{if } |k/n - c_j| \leq \rho \text{ for some } j, \\ \bar{d}, & \text{otherwise.} \end{cases} \tag{18}$$

By construction, there is at most one such j.

For a state ω, set

$$v_{n,k}(\omega) := \int_A u^p(a,\omega)\, d_n(k)(da),$$

and, for $p \in [0,1]$, define

$$W_{n,\omega}(p) := \sum_{k=0}^{n} b_{n,p}(k) v_{n,k}(\omega). \tag{19}$$

This is the decision-maker's conditional expected payoff when the reports are independent Bernoulli random variables with parameter p.

Let

$$x_j^- := c_j - 3\rho \qquad \text{and} \qquad x_j^+ := c_j + 3\rho.$$

Chebyshev's inequality and $\mathrm{Var}(\mathrm{Bin}(n,p)/n) \le 1/(4n)$ imply, uniformly in ω and j,

$$\begin{aligned} &|W_{n,\omega}(c_j) - u^p(a_j,\omega)| \le e_n, \\ &\left|W_{n,\omega}(x_j^\pm) - \bar{u}(\omega)\right| \le e_n, \qquad\qquad e_n := \frac{B}{4n\rho^2} \longrightarrow 0. \end{aligned} \tag{20}$$

At c_j, the report fraction leaves the interval around c_j only if it differs from its mean by at least ρ, so Chebyshev's inequality bounds this probability by $1/(4n\rho^2)$. At $x_j^\pm$, reaching any such interval requires a deviation of at least 2ρ, so the probability is at most $1/(16n\rho^2)$. Since two action payoffs differ by at most B, both errors are bounded by e_n.

Suppose first that $g(\omega) > 2e_n$, and let $j^*(\omega)$ be the smallest index satisfying

$$u^p(a_{j^*(\omega)},\omega) = M_K(\omega).$$

By (20),

$$\begin{aligned} &W_{n,\omega}(c_{j^*}) \ge M_K(\omega) - e_n, \\ &W_{n,\omega}(x_{j^*}^\pm) \le \bar{u}(\omega) + e_n = M_K(\omega) - g(\omega) + e_n < M_K(\omega) - e_n. \end{aligned}$$

Therefore, $W_{n,\omega}$ attains its maximum over $[x_{j^*}^-, x_{j^*}^+]$ at an interior point. Let $p_{n,\omega}^*$ be the smallest maximizer over this interval. Because there are finitely many possible values of $j^*(\omega)$ and $W_{n,\omega}(p)$ is jointly continuous in (p,ω), both $j^*(\omega)$ and $p_{n,\omega}^*$ are measurable in ω.

Since (19) is a polynomial in p, its derivative vanishes at $p_{n,\omega}^*$. Differentiating the binomial

polynomial gives

$$\frac{d}{dp}W_{n,\omega}(p) = n\sum_{k=0}^{n-1} b_{n-1,p}(k)\big[v_{n,k+1}(\omega) - v_{n,k}(\omega)\big] =: n\Delta_{n,\omega}(p), \tag{21}$$

where $\Delta_{n,\omega}(p)$ is defined as an individual's gross expected benefit from reporting when the other $n-1$ individuals report with probability p. Hence,

$$\Delta_{n,\omega}(p^*_{n,\omega}) = 0.$$

Because $p^*_{n,\omega} \in (s_-, s_+)$, it can be written as

$$p^*_{n,\omega} = s_- + \alpha\gamma_{n,\omega} \qquad \text{for some } \gamma_{n,\omega} \in (0,1).$$

Let types with $\nu_i < 0$ report, let types with $\nu_i > 0$ not report, and let zero-cost types report with probability $\gamma_{n,\omega}$. This strategy induces $p^*_{n,\omega}$. Since the gross reporting benefit is zero, the net benefit from reporting is $-\nu_i$, so every type is playing a best response. This strategy profile is therefore a symmetric equilibrium, and its conditional payoff is at least $M_K(\omega) - e_n$. Moreover, because $p^*_{n,\omega}$ is measurable, so is $\gamma_{n,\omega}$, and hence this equilibrium strategy is measurable in ω.

Now consider states with $g(\omega) \le 2e_n$. For every j,

$$0 \le M_K(\omega) - u^p(a_j,\omega) \le \sum_{\ell=1}^{K}\big[M_K(\omega) - u^p(a_\ell,\omega)\big] = Kg(\omega) \le 2Ke_n.$$

Thus every action distribution induced by d_n gives the decision-maker an expected payoff in state ω of at least $M_K(\omega) - 2Ke_n$. For states with $g(\omega) \le 2e_n$, take the measurable symmetric equilibrium from Lemma C2; for states with $g(\omega) > 2e_n$, take the equilibrium constructed above.

Since g is continuous, the sets on which these two constructions are used are measurable. Combining the two measurable statewise strategies therefore gives a symmetric Bayesian Nash equilibrium under d_n. The decision-maker's expected payoff in every state is at least

$$M_K(\omega) - 2Ke_n.$$

Using (17) and taking expectations gives

$$\Pi(n) \ge \Pi^{FB} - \xi - 2Ke_n.$$

Letting $n \to \infty$ and then $\xi \downarrow 0$ yields

$$\liminf_{n\to\infty} \Pi(n) \geq \Pi^{FB}.$$

Since $\Pi(n) \leq \Pi^{FB}$ for every n, Statement 4 follows. □

Proof of Proposition 3. The Volunteer's Dilemma is a special case of the generalized environment. Consequently, Statements 1 and 2, the first claim in Statement 3, and Statement 4 follow directly from Statements 1–4, respectively, of Proposition 4. It remains only to establish the assertions concerning threshold decision rules in Statement 3.

Write $\pi := \Pr(\omega = 1)$ and set $t := F'(0) > 0$. The maintained assumptions imply $\pi \in (0,1)$, $H > 0$, $L < 0$, and $\pi H - (1-\pi)|L| < 0$. Lemma A1 gives

$$\lim_{n\to\infty} \Pi^T(n) = \Pi^{SQ} + V(t),$$

so the limit exists. Lemma A2 gives $V(t) > 0$ if and only if $t > \overline{F}_0$, where $\overline{F}_0 \in (0,\infty)$ depends only on π, H, L, and $F(0)$. Therefore

$$\liminf_{n\to\infty} \Pi^T(n) > \Pi^{SQ} \quad \Longleftrightarrow \quad F'(0) > \overline{F}_0,$$

which proves the remaining claims in Statement 3. □

B Microfoundations

All information-transmission results use Assumption 1 only through Lemma 1. Assumption 1 has two interpretations: (i) some individuals intrinsically enjoy reporting or (ii) behavior follows a QRE. This appendix drops Assumption 1 and gives two additional microfoundations—behavioral types and noisy communication—that imply Lemma 1 and therefore preserve all subsequent results under reinterpretations of $F(\cdot)$ specific to those analyses.[12] It then considers biased individuals, whose presence does not imply Lemma 1 but may nonetheless generate information-aggregation failures in large groups.

[12]Specifically, with c denoting the gross benefit of raising r_i from 0 to 1, the reinterpreted $F(c)$ is $\epsilon_1 + (1 - \epsilon_1 - \epsilon_0)F(c)$ with behavioral types and $(1-p) + (2p-1)F((2p-1)c)$ with noisy communication, where F on the right-hand sides is the distribution of ν_i and the parameters are defined in the corresponding subsections.

B.1 Behavioral Types

In addition to the rational types described in the model, there are two behavioral types: one always reports ($r_i = 1$), and the other never reports ($r_i = 0$). Let ϵ_1 denote the probability that an individual is a behavioral type who always reports 1, and let ϵ_0 denote the probability that an individual is a behavioral type who always reports 0. Therefore, with probability $1 - \epsilon_1 - \epsilon_0$, an individual is a rational type as described in the model. Lemma 1 holds under this microfoundation upon taking $\epsilon_* = \min\{\epsilon_1, \epsilon_0\} > 0$.

B.2 Noisy Communication

Motivated by Boudreau et al. (2023), consider the following adapted model. The individual's and the decision-maker's utility functions remain as in the main model. However, now the individual chooses $\tilde{r}_i \in \{0, 1\}$ and receives utility $u^p(a, \omega) - \nu_i \tilde{r}_i$. The decision-maker observes $\sum_i r_i$, where $r_i = \tilde{r}_i$ with probability $p \in (1/2, 1)$ and $r_i = 1 - \tilde{r}_i$ with probability $1 - p$. The decision-maker can commit to any decision rule as a function of $\sum_i r_i$. If p were equal to 1, then this would coincide with the main model, where an individual's report is transmitted without noise. In contrast, if p were equal to $1/2$, then the decision-maker would observe reports that are independent of what an individual reports. One can now calculate

$$1 - p = \Pr(r_i = 1 \mid \tilde{r}_i = 0, \omega) \leq \Pr(r_i = 1 \mid \omega) \leq \Pr(r_i = 1 \mid \tilde{r}_i = 1, \omega) = p \tag{22}$$

$$1 - p = \Pr(r_i = 0 \mid \tilde{r}_i = 1, \omega) \leq \Pr(r_i = 0 \mid \omega) \leq \Pr(r_i = 0 \mid \tilde{r}_i = 0, \omega) = p, \tag{23}$$

where these inequalities hold for every ω. Therefore, with $\epsilon_* = 1 - p > 0$, Lemma 1 holds with noisy communication as defined above. While the decision-maker conditions on the observed reports r_i, an individual pays ν_i for the intended report $\tilde{r}_i$, and switching $\tilde{r}_i$ from 0 to 1 raises $\Pr(r_i = 1)$ from $1 - p$ to p. Hence the gross benefit of raising one's intended report equals $(2p - 1)$ times the gross benefit of raising one's observed report. By Lemma 1, the observed reporting probabilities lie in $[1 - p,\, p] \subset [\epsilon_*,\, 1 - \epsilon_*]$, so the pivotality bound of Lemma 2 applies to the observed count, and the latter benefit is at most $M/\sqrt{n}$. Therefore, the benefit from reporting (excluding costs) is at most $(2p - 1)M/\sqrt{n} < M/\sqrt{n}$. Because this incentive stays within the cutoff band of Lemma 2, the same argument yields the upper bounds on the decision-maker's asymptotic payoff in Proposition 3. Finally, Statement 4 (i.e., that the decision-maker can achieve her first-best payoff when $F(\cdot)$ has a mass point at zero) is also preserved: a zero-cost type is indifferent exactly when its gross benefit vanishes, which is unaffected by the $(2p - 1)$ scaling, so such types can still mix as in the construction of Proposition 4. Therefore, whenever F has an atom at zero, $\lim_{n\to\infty} \Pi(n) = \Pi^{FB}$ under

noisy communication as well.

Finally, the simplest interpretation of p is the probability of correct transmission under intentional garbling of an individual's report. In more applied contexts, one could also view $1-p$ as the probability of a misinterpretation arising from a language barrier or other communication failure. While formal modeling of such miscommunication is sparse (for the reasons described in Lipman, 2025), miscommunications such as those in Cremer et al. (2007) or Blume et al. (2007) could serve as a microfoundation.

B.3 Biased Individuals

In this subsection, I specialize to binary actions and states, as in the Volunteer's Dilemma. I assume that, in addition to the rational types described in the model, there exist upward-biased individuals with utility a and downward-biased individuals with utility $-a$. I assume an individual is upward biased with probability $p_u \in (0,1)$, downward biased with probability $p_d \in (0, 1-p_u)$, and unbiased (sharing the decision-maker's preferences) with probability $1-p_u-p_d > 0$. I begin with a result for monotone decision rules, where a monotone decision rule is one in which the action is monotone in the number of reports. The class of monotone decision rules strictly contains the class of threshold rules considered in the paper.

Proposition B1. *If $p_u, p_d > 0$, then for any monotone and nonconstant decision rule there exists an $\epsilon_* > 0$ such that in any equilibrium in which the individuals do not play weakly dominated actions,* $\Pr(r_i = 1 \mid \omega) \in [\epsilon_*, 1-\epsilon_*]$.

Proof of Proposition B1. Without loss of generality, consider a weakly increasing decision rule; the decreasing case is symmetric. With a weakly increasing decision rule, the downward-biased individual has a weakly dominant strategy to choose $r_i = 0$, whereas the upward-biased individual has a weakly dominant strategy to choose $r_i = 1$. Since the rule is non-constant, some k has $\mathbb{E}_{a\sim d(k+1)}(a) > \mathbb{E}_{a\sim d(k)}(a)$, so $r_i = 0$ is weakly dominated for upward-biased individuals, and symmetrically for downward-biased individuals. Therefore, setting $\epsilon_* = \min\{p_u, p_d\}$ completes the proof. □

Given this result, the upper bounds in Proposition 3 in the main text continue to hold when $\Pi(n)$ is redefined as the supremum over all monotone decision rules and all equilibria in which the individuals do not play weakly dominated actions. I now show that a very natural class of decision rules is monotone. When the decision-maker cannot commit, call a rule ex post efficient if its action is optimal given the posterior induced by reports.

Remark 1. *Any ex post efficient decision rule is monotone on path.*

Proof of Remark 1. Let q^1, q^0 denote the reporting probabilities when $\omega = 1$ and $\omega = 0$, respectively. If $q^1 = q^0$, then the decision-maker's beliefs will be independent of the number of messages received. Therefore, the decision-maker's optimal action is her prior action following any number of reports, which is monotone. Thus, assume $q^1 \neq q^0$. If both q^1 and q^0 lie on the boundary of $[0, 1]$ and $q^1 \neq q^0$, then on path the reports perfectly reveal the state, implying that the ex post efficient decision rule is monotone on path. If only one of q^1 and q^0 lies in $\{0, 1\}$, the likelihood ratio is weakly monotone on the on-path support, so the conclusion follows directly. Thus, suppose $q^1, q^0 \in (0, 1)$ are interior and $q^1 \neq q^0$. Suppose the decision-maker observes the total number of reports $k = \sum_i r_i$ and updates her beliefs. Because $q^1 \neq q^0$, the binomial distributions $\text{Bin}(n, q^1)$ and $\text{Bin}(n, q^0)$ satisfy the monotone likelihood-ratio property. The likelihood ratio is

$$\Lambda(k) = \frac{\Pr(k \mid \omega = 1)}{\Pr(k \mid \omega = 0)} = \left(\frac{q^1}{q^0}\right)^k \left(\frac{1 - q^1}{1 - q^0}\right)^{n-k}.$$

Because $\frac{q^1(1-q^0)}{q^0(1-q^1)} \neq 1$, $\Lambda(k)$ is strictly monotone in k. Because the optimal action is increasing in the posterior probability of the state $\omega = 1$, the ex post efficient action is therefore monotone in the number of reports. □

Finally, even under nonmonotone decision rules, the reporting probability remains intermediate if biased individuals break indifference following a natural tie-breaking rule.

Proposition B2. *Suppose $p_u, p_d > 0$ and that if the upward-biased individual (respectively, downward-biased individual) is indifferent between reporting and not, the individual chooses to report (respectively, not report). For any decision rule, there exists an $\epsilon_* > 0$ such that in any equilibrium $\Pr(r_i = 1 \mid \omega) \in [\epsilon_*, 1 - \epsilon_*]$.*

Proof of Proposition B2. For any decision rule, define $X := \sum_{j \neq i} r_j$, the number of reports submitted by the other $n - 1$ individuals, and define

$$\Delta_n(\omega) := \sum_{k=0}^{n-1} \Pr(X = k \mid \omega) \left(\mathbb{E}_{a \sim d(k+1)}(a) - \mathbb{E}_{a \sim d(k)}(a)\right), \tag{24}$$

which captures whether individual i's report increases or decreases the action in expectation.

If $\Delta_n(\omega) \neq 0$, then the upward-biased individuals choose $r_i = \mathbf{1}_{\Delta_n(\omega) > 0}$, and the downward-biased individuals choose $r_i = \mathbf{1}_{\Delta_n(\omega) < 0}$. In contrast, if $\Delta_n(\omega) = 0$, then the tie-breaking rule imposed implies that $r_i = 1$ for the upward-biased individuals and $r_i = 0$ for the downward-biased individuals. Therefore, in either case, we can set $\epsilon_* = \min\{p_u, p_d\}$. □

Therefore, even when allowing for nonmonotone decision rules, given this tie-breaking assumption, the bounded reporting probabilities imply that the unbiased individuals face vanishing expected marginal benefits from reporting. As a result, if the cost distribution of these unbiased individuals is bounded away from zero, any information aggregation in sufficiently large groups must come from the biased individuals with directional preferences. Such aggregation is not impossible: for instance, relaxing the tie-breaking assumption, consider the case where all costs are nonnegative and a mechanism under which $a = 1 \iff \sum_i r_i \geq T_n$, with $T_n > 1$, has the following equilibrium: when $\omega = 0$, no individual reports, and when $\omega = 1$, the upward-biased individuals always report. This is an equilibrium because, when $\omega = 0$, no individual has an incentive to report. However, this equilibrium relies on (i) the upward-biased individuals using a weakly dominated strategy and (ii) their equilibrium strategies depending on the realization of ω, which is not payoff relevant for them.

C Mathematical Appendix

Lemma C1. *Let $P_n = \mathrm{Bin}(n, p_n)$ and $Q_n = \mathrm{Bin}(n, q_n)$, where $p_n, q_n \in [\lambda, 1-\lambda]$ for some $\lambda \in (0, 1/2]$. If $n(p_n - q_n)^2 \to 0$, then $\mathrm{TV}(P_n, Q_n) \to 0$. Moreover, if $n(p_n - q_n)^2 \leq C < \infty$, then, for all sufficiently large n,*

$$\mathrm{TV}(P_n, Q_n) \leq \sqrt{1 - \exp(-C/\lambda)} < 1. \tag{25}$$

Proof of Lemma C1. Define

$$\mathcal{A}_n = \sum_{k=0}^{n} \sqrt{P_n(k) Q_n(k)} = \left(\sqrt{p_n q_n} + \sqrt{(1-p_n)(1-q_n)} \right)^n =: b_n^n, \tag{26}$$

$$1 - b_n = \frac{1}{2} \left[(\sqrt{p_n} - \sqrt{q_n})^2 + (\sqrt{1-p_n} - \sqrt{1-q_n})^2 \right] \leq \frac{(p_n - q_n)^2}{4\lambda}. \tag{27}$$

For any probability mass functions P, Q, Cauchy–Schwarz yields

$$\mathrm{TV}(P, Q) \leq \sqrt{1 - \left(\sum_x \sqrt{P(x) Q(x)} \right)^2}. \tag{28}$$

If $n(p_n - q_n)^2 \to 0$, then $1 - \mathcal{A}_n \leq n(1 - b_n)$, so

$$\mathrm{TV}(P_n, Q_n) \leq \sqrt{2n(1-b_n)} \leq \sqrt{\frac{n(p_n - q_n)^2}{2\lambda}} \longrightarrow 0. \tag{29}$$

If $n(p_n - q_n)^2 \le C$, then for all sufficiently large n, $1 - b_n \le 1/2$ and $\log b_n \ge -2(1 - b_n)$. Hence $\mathcal{A}_n \ge \exp[-C/(2\lambda)]$, and (28) gives (25). □

Lemma C2 (Symmetric-Equilibrium Existence)**.** *For every n and every symmetric decision rule d, there exists a symmetric Bayesian Nash equilibrium.*

Proof of Lemma C2. Fix n and d, and let $v_k(\omega) := \int_A u^p(a, \omega)\, d(k)(da)$ be the decision-maker's expected payoff in state ω when k reports are made. Each v_k is continuous because u^p is uniformly continuous on the compact set $A \times \Omega$. If each other individual reports independently with probability q, an individual's gross expected benefit from reporting is

$$\Delta_\omega(q) := \sum_{k=0}^{n-1} \binom{n-1}{k} q^k (1-q)^{n-1-k} \big(v_{k+1}(\omega) - v_k(\omega)\big),$$

which is therefore jointly continuous in (ω, q). Types with $\nu_i < \Delta_\omega(q)$ strictly prefer to report, types with $\nu_i > \Delta_\omega(q)$ strictly prefer not to, and types with $\nu_i = \Delta_\omega(q)$ are indifferent. Since $F(c^-) = \Pr(\nu_i < c)$, best responses to q can therefore induce the reporting probability q itself if and only if

$$F(\Delta_\omega(q)^-) \le q \le F(\Delta_\omega(q)). \tag{30}$$

Let $E := \{(\omega, q) \in \Omega \times [0,1] : F(\Delta_\omega(q)^-) \le q\}$, and for each ω let $E_\omega := \{q \in [0,1] : (\omega, q) \in E\}$. Because $c \mapsto F(c^-)$ is nondecreasing and left-continuous, it is lower semicontinuous; since $\Delta_\omega(q)$ is continuous, E is closed, and hence so is each E_ω. Since $1 \in E_\omega$ for every ω, $q_\omega := \min E_\omega$ exists and satisfies the first inequality in (30). The second inequality is immediate if $q_\omega = 0$. If $q_\omega > 0$, every $q \in [0, q_\omega)$ lies outside E_ω, so $q < F(\Delta_\omega(q)^-) \le F(\Delta_\omega(q))$. Since F is nondecreasing and right-continuous, it is upper semicontinuous, and letting $q \uparrow q_\omega$ gives $q_\omega \le F(\Delta_\omega(q_\omega))$. Moreover, if $\omega_j \to \omega$, then along a subsequence $q_{\omega_j} \to \liminf_j q_{\omega_j}$; because E is closed, this limit lies in E_ω and is therefore at least q_ω. Thus $\omega \mapsto q_\omega$ is lower semicontinuous and hence Borel measurable.

Let $c_\omega := \Delta_\omega(q_\omega)$. In state ω, let types with $\nu_i < c_\omega$ report, types with $\nu_i > c_\omega$ not report, and types with $\nu_i = c_\omega$ report with probability $\alpha_\omega := \big(q_\omega - F(c_\omega^-)\big)/\big(F(c_\omega) - F(c_\omega^-)\big)$ if $F(c_\omega) > F(c_\omega^-)$, and $\alpha_\omega := 0$ otherwise. By (30), $\alpha_\omega \in [0,1]$, and this strategy induces the reporting probability q_ω, so the gross benefit from reporting is c_ω, and every type is best responding in every state. Finally, since $\Delta_\omega(q)$ is continuous and both F and $c \mapsto F(c^-)$ are monotone, c_ω and α_ω are Borel measurable, so the strategy is jointly measurable in (ω, ν_i) and constitutes a symmetric Bayesian Nash equilibrium. □

Lemma C3 (Uniform Binomial Bounds and Local Limits)**.** *Fix $\lambda \in (0, 1/2]$. There exists $C_\lambda < \infty$ such that, for all $m \geq 1$, $p \in [\lambda, 1-\lambda]$, and $X_{m,p} \sim \mathrm{Bin}(m, p)$,*

$$\max_{k \in \{0,\dots,m\}} \Pr(X_{m,p} = k) \leq \frac{C_\lambda}{\sqrt{m}}. \tag{31}$$

Moreover, fix $s \in (0,1)$ and $K, L_0 < \infty$, and let $k_n(t) := \lfloor ns + t\sqrt{n} \rfloor$. Uniformly over $|x|, |t| \leq K$ and integers $|\ell| \leq L_0$,

$$\sqrt{n}\Pr\left(\mathrm{Bin}\left(n+\ell, s + \frac{x}{\sqrt{n}}\right) = k_n(t)\right) \to \frac{1}{\sqrt{s(1-s)}}\varphi\left(\frac{t-x}{\sqrt{s(1-s)}}\right), \tag{32}$$

$$\Pr\left(\mathrm{Bin}\left(n+\ell, s + \frac{x}{\sqrt{n}}\right) \geq k_n(t)\right) \to 1 - \Phi\left(\frac{t-x}{\sqrt{s(1-s)}}\right). \tag{33}$$

Proof of Lemma C3. Write $X_{m,p} = \sum_{j=1}^m \xi_j$, where $\xi_1, \dots, \xi_m$ are independent Bernoulli(p) random variables, and let $\sigma_{m,p} := \sqrt{mp(1-p)}$ be the standard deviation of $X_{m,p}$. Since $p(1-p) \geq \lambda(1-\lambda)$ for every $p \in [\lambda, 1-\lambda]$, we have $\sigma_{m,p} \geq \sqrt{m\,\lambda(1-\lambda)}$. Each ξ_j has variance $p(1-p)$ and third absolute central moment $\mathbb{E}|\xi_j - p|^3 = p(1-p)\big(p^2 + (1-p)^2\big)$, so

$$\frac{\mathbb{E}|\xi_j - p|^3}{\big(p(1-p)\big)^{3/2}} = \frac{p^2 + (1-p)^2}{\sqrt{p(1-p)}} \leq \frac{1}{\sqrt{\lambda(1-\lambda)}},$$

where the inequality uses $p^2 + (1-p)^2 \leq 1$. Let C_{BE} denote the universal constant in the Berry–Esseen Theorem and set $B_\lambda := C_{\mathrm{BE}}/\sqrt{\lambda(1-\lambda)}$. The Berry–Esseen Theorem then gives

$$\sup_{z \in \mathbb{R}} \Big| \Pr\big(X_{m,p} \leq mp + z\,\sigma_{m,p}\big) - \Phi(z) \Big| \leq \frac{B_\lambda}{\sqrt{m}}, \qquad m \geq 1,\ p \in [\lambda, 1-\lambda]. \tag{34}$$

We first prove (31). For any integer k, subtracting (34) at $z = (k-1-mp)/\sigma_{m,p}$ from (34) at $z = (k-mp)/\sigma_{m,p}$,

$$\Pr(X_{m,p} = k) \leq \Phi\Big(\tfrac{k-mp}{\sigma_{m,p}}\Big) - \Phi\Big(\tfrac{k-1-mp}{\sigma_{m,p}}\Big) + \frac{2B_\lambda}{\sqrt{m}} \leq \frac{1}{\sqrt{2\pi}\,\sigma_{m,p}} + \frac{2B_\lambda}{\sqrt{m}} \leq \frac{C_\lambda}{\sqrt{m}},$$

where the middle step uses $\Phi' = \varphi \leq 1/\sqrt{2\pi}$ over an interval of length $1/\sigma_{m,p}$, and the last step uses $\sigma_{m,p} \geq \sqrt{m\,\lambda(1-\lambda)}$ with $C_\lambda := 1/\sqrt{2\pi\,\lambda(1-\lambda)} + 2B_\lambda$. Taking the maximum over $k \in \{0, \dots, m\}$ gives (31).

For the remaining claims, fix $\lambda_0 \in (0, 1/2)$ with $s \in (\lambda_0, 1-\lambda_0)$, and take $n > L_0$ large enough that $p := s + x/\sqrt{n} \in [\lambda_0, 1-\lambda_0]$ for every $|x| \leq K$. Write $q = 1-p$, $m = n + \ell$,

$X \sim \mathrm{Bin}(m,p)$, and

$$z_n := \frac{k_n(t) - mp}{\sigma_{m,p}}.$$

Because $|k_n(t) - (ns + t\sqrt{n})| \le 1$ and $mp = ns + x\sqrt{n} + \ell s + \ell x/\sqrt{n}$, we have $|k_n(t) - mp - (t - x)\sqrt{n}| \le 1 + L_0(1+K)$; dividing by $\sigma_{m,p} = \sqrt{n}\,\sqrt{(m/n)\,pq}$ and using $(m/n)\,pq \to s(1-s)$, uniformly over the parameter set,

$$z_n \to \frac{t-x}{\sqrt{s(1-s)}} \qquad \text{and} \qquad \frac{\sqrt{n}}{\sigma_{m,p}} \to \frac{1}{\sqrt{s(1-s)}}. \tag{35}$$

In particular, $|z_n| \le C_z$ for some $C_z < \infty$ and all large n.

For (33), apply (34) with λ_0 in place of λ at $z = z_n - 1/\sigma_{m,p}$:

$$\Big|\Pr\big(X \le k_n(t) - 1\big) - \Phi\big(z_n - \tfrac{1}{\sigma_{m,p}}\big)\Big| \le \frac{B_{\lambda_0}}{\sqrt{m}}.$$

By (35), uniform continuity of Φ, and $B_{\lambda_0}/\sqrt{m} \to 0$, the following convergence is uniform over the parameter set:

$$\Pr\big(X \ge k_n(t)\big) = 1 - \Pr\big(X \le k_n(t) - 1\big) \to 1 - \Phi\Big(\frac{t-x}{\sqrt{s(1-s)}}\Big).$$

For (32), set $k = k_n(t)$; for n large, $k, m-k \ge m\lambda_0/2$. Put

$$w_1 := z_n\sqrt{\tfrac{q}{mp}}, \qquad w_2 := -\,z_n\sqrt{\tfrac{p}{mq}},$$

so that $k = mp(1+w_1)$, $m-k = mq(1+w_2)$, and $|w_1|, |w_2| \le C_z/\sqrt{m\lambda_0}$. By Stirling's approximation $j!/(\sqrt{2\pi j}\,(j/e)^j) \to 1$ applied to $m!, k!, (m-k)!$,

$$\Pr(X = k) = \binom{m}{k} p^k q^{m-k} = \sqrt{\frac{m}{2\pi k(m-k)}}\,\Big(\frac{mp}{k}\Big)^k \Big(\frac{mq}{m-k}\Big)^{m-k} \kappa_n,$$

where $\kappa_n \to 1$ uniformly, since the Stirling remainders tend to one and $\min\{m, k, m-k\} \to \infty$ uniformly. Using $\big|(1+w)\log(1+w) - w - \frac{w^2}{2}\big| \le |w|^3$ for $|w| \le \frac{1}{2}$ (which holds here for n large),

$$\log\Big[\Big(\tfrac{mp}{k}\Big)^k \Big(\tfrac{mq}{m-k}\Big)^{m-k}\Big] = -mp(1+w_1)\log(1+w_1) - mq(1+w_2)\log(1+w_2) = -\frac{z_n^2}{2} + \theta_n,$$

because $mp\,w_1 + mq\,w_2 = 0$ and $mp\,w_1^2 + mq\,w_2^2 = z_n^2$, while the remainder obeys $|\theta_n| \le$

$mp|w_1|^3 + mq|w_2|^3 \le 2C_z^3/\sqrt{m\lambda_0} \to 0$ uniformly. Since $k(m-k) = m^2pq\,(1+w_1)(1+w_2)$,

$$\sqrt{\frac{m}{2\pi k(m-k)}} = \frac{1}{\sigma_{m,p}\sqrt{2\pi}} \cdot \frac{1}{\sqrt{(1+w_1)(1+w_2)}}.$$

Combining the last three equations,

$$\Pr(X = k_n(t)) = \frac{1}{\sigma_{m,p}}\,\varphi(z_n) \cdot \frac{\kappa_n\, e^{\theta_n}}{\sqrt{(1+w_1)(1+w_2)}},$$

and the final factor converges to one uniformly, as $w_1, w_2, \theta_n \to 0$. Multiplying by $\sqrt{n}$ and using (35) together with the continuity of φ, all uniformly,

$$\sqrt{n}\,\Pr(X = k_n(t)) = \frac{\sqrt{n}}{\sigma_{m,p}}\,\varphi(z_n) \cdot \big(1 + o(1)\big) \to \frac{1}{\sqrt{s(1-s)}}\,\varphi\Big(\frac{t-x}{\sqrt{s(1-s)}}\Big),$$

which is (32). □

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Online Appendix

D Extensions

This section contains the formalisms for the extensions discussed in Subsection 5.2 in the paper.

D.1 State-Dependent Reporting Costs

Consider the binary Volunteer's Dilemma of Section 4, except that the reporting cost ν_i is drawn from a state-dependent distribution F_ω, $\omega \in \{0,1\}$. I show that the upper bounds on the limiting payoffs continue to hold, provided that $F_1(0) = F_0(0)$. This assumption states that the proportion of individuals with nonpositive reporting costs is the same in both states.

Proposition D1 (State-Dependent Reporting Costs)**.** *Let each F_ω satisfy the analogue of Assumption 1, and suppose $F_1(0) = F_0(0) =: s$.*

1. *If there exists $\delta > 0$ such that $F_\omega(\delta) - F_\omega(-\delta) = 0$ for both $\omega \in \{0,1\}$, then there exists $n^* \in \mathbb{N}$ such that*
$$\Pi(n) = \Pi^{SQ} \qquad \text{for all } n > n^*.$$

2. *If F_1 and F_0 are differentiable at 0 with $F_1'(0) = F_0'(0) = 0$, then*
$$\lim_{n\to\infty} \Pi(n) = \Pi^{SQ}.$$

3. *If F_1 and F_0 are differentiable at 0, then there exists $\beta \in (0,1)$ such that*
$$\limsup_{n\to\infty} \Pi(n) \leq \beta\Pi^{SQ} + (1-\beta)\Pi^{FB} < \Pi^{FB}.$$

If instead $F_1(0) \neq F_0(0)$, then $\lim_{n\to\infty} \Pi(n) = \Pi^{FB}$.

Proof of Proposition D1. Fix n, a symmetric decision rule d, and a symmetric equilibrium e. Let $\epsilon_* > 0$ be such that the conclusion of Lemma 1 holds for both F_1 and F_0, let M be the corresponding constant in Lemma 2, and write
$$p_{n,\omega} := \Pr(r_i = 1 \mid \omega), \qquad h_n := \frac{M}{\sqrt{n}}.$$

For notational convenience, the dependence on d and e is suppressed. Lemma 2 and monotonicity of F_ω imply, uniformly over d and e,

$$F_\omega\big((-h_n)^-\big) \le p_{n,\omega} \le F_\omega(h_n), \qquad \omega \in \{0,1\}. \tag{D.1}$$

Define

$$p_n^- := \max\Big\{\min_\omega F_\omega\big((-h_n)^-\big), \epsilon_*\Big\}, \qquad p_n^+ := \min\Big\{\max_\omega F_\omega(h_n), 1-\epsilon_*\Big\}.$$

By (D.1) and Lemma 1, $p_{n,\omega} \in [p_n^-, p_n^+] \subseteq [\epsilon_*, 1-\epsilon_*]$ uniformly over ω, d, e. The proof of Statements 1–3 of Proposition 4 uses only that the reporting probabilities lie in $[p_n^-, p_n^+]$, so it gives

$$\Pi^{SQ} \le \Pi(n) \le \beta_n \Pi^{SQ} + (1-\beta_n)\Pi^{FB}, \qquad \beta_n := 1 - \mathrm{TV}\big(\mathrm{Bin}(n, p_n^-), \mathrm{Bin}(n, p_n^+)\big).$$

For Statement 1, choose n^* so that $h_n < \delta$ for every $n > n^*$. Each F_ω is then constant on $[-h_n, h_n]$, with value s, so $p_n^- = p_n^+$, $\beta_n = 1$, and $\Pi(n) = \Pi^{SQ}$.

For Statements 2 and 3, $F_\omega((-h_n)^-) \ge F_\omega(-2h_n)$, $F_1(0) = F_0(0)$, and differentiability at zero imply

$$\sqrt{n}\,(p_n^+ - p_n^-) \le \sqrt{n}\Big(\max_\omega F_\omega(h_n) - \min_\omega F_\omega(-2h_n)\Big) \longrightarrow 3M \max\{F_1'(0), F_0'(0)\}.$$

If $F_1'(0) = F_0'(0) = 0$, then $n(p_n^+ - p_n^-)^2 \to 0$, so Lemma C1 implies that $\beta_n \to 1$ and therefore that $\Pi(n) \to \Pi^{SQ}$, which proves Statement 2. In general, $n(p_n^+ - p_n^-)^2$ is bounded, so Lemma C1 gives $\beta \in (0,1)$ with $\beta_n \ge \beta$ for all sufficiently large n. As $\Pi^{FB} > \Pi^{SQ}$, it follows that

$$\limsup_{n\to\infty} \Pi(n) \le \beta\Pi^{SQ} + (1-\beta)\Pi^{FB} < \Pi^{FB},$$

which proves Statement 3.

Final statement ($F_1(0) \ne F_0(0)$). Suppose first that $F_1(0) > F_0(0)$. Choose $\tau \in \big(F_0(0), F_1(0)\big)$ and integers T_n such that $T_n/n \to \tau$; therefore, for all sufficiently large n, $T_n \in \{1, \dots, n\}$. Consider the decision rule

$$a = 1 \quad \Longleftrightarrow \quad \sum_{i=1}^n r_i \ge T_n.$$

By Lemma 1, the probability any individual is pivotal is nonzero. A pivotal report triggers an investigation, which yields $H > 0$ in state 1 and $L < 0$ in state 0, so the gross benefit

from reporting is positive in state 1 and negative in state 0. Hence all types with $\nu_i \leq 0$ report in state 1, and all types with $\nu_i \geq 0$ refrain in state 0, so

$$p_{n,1} \geq F_1(0) > \tau > F_0(0) \geq p_{n,0}.$$

Thus there exists $\eta > 0$, independent of n, such that

$$p_{n,1} \geq \tau + \eta \qquad \text{and} \qquad p_{n,0} \leq \tau - \eta$$

for all sufficiently large n. Since

$$\operatorname{Var}\left(\frac{1}{n}\sum_{i=1}^{n} r_i \mid \omega\right) \leq \frac{1}{4n},$$

Chebyshev's inequality implies

$$\Pr(a = w \mid \omega = w) \longrightarrow 1 \qquad \text{for each } w \in \{0, 1\}.$$

If instead $F_1(0) < F_0(0)$, the same argument works with a decision rule under which $a = 1 \iff \sum_i r_i < T_n$. In either case, an equilibrium exists under this rule: the fixed-point argument in Lemma C2 applies in each state with F_ω in place of F, and measurability is immediate because there are only two states. Hence $\lim_{n\to\infty} \Pi(n) = \Pi^{FB}$. □

D.2 Worst-Case Equilibrium Selection

The upper bounds in Proposition 3 continue to hold under worst-case equilibrium selection: allowing best-case selection can only raise the decision-maker's payoff. The only new arguments below therefore concern lower bounds. Define

$$\underline{\Pi}_n := \sup_{d \in D(n)} \inf_{e \in \mathcal{E}(d)} \mathbb{E}_{e,d}\big[u^p(a, \omega)\big], \qquad \underline{\Pi}_n^T := \sup_{d \in D^T(n)} \inf_{e \in \mathcal{E}(d)} \mathbb{E}_{e,d}\big[u^p(a, \omega)\big]. \tag{D.2}$$

Proposition D2 (Worst-Case Equilibrium Selection)**.** *Recall that $F(\cdot)$ denotes the distribution of reporting costs. The following hold:*

1. *If there exists a $\delta > 0$ such that $F(\delta) - F(-\delta) = 0$, then there exists $n^* \in \mathbb{N}$ such that, for every $n > n^*$, $\underline{\Pi}_n = \Pi^{SQ}$.*

2. *Suppose F is differentiable at zero. If $F'(0) = 0$, then $\lim_{n\to\infty} \underline{\Pi}_n = \Pi^{SQ}$.*

3. *Suppose F is differentiable at zero and $F'(0) > 0$. Then $\limsup_{n\to\infty} \underline{\Pi}_n < \Pi^{FB}$. If, in*

addition, F is continuously differentiable on an open neighborhood of zero, then, writing $\pi := \Pr(\omega = 1)$, there exists a constant $\underline{F}_0 \in (0, \infty)$, depending only on π, H, L, and $F(0)$, such that

$$F'(0) > \underline{F}_0 \quad \Longrightarrow \quad \liminf_{n\to\infty} \underline{\Pi}_n^T > \Pi^{SQ}.$$

4. If $\lim_{\delta\downarrow 0}[F(\delta) - F(-\delta)] > 0$, then $\lim_{n\to\infty} \underline{\Pi}_n^T = \Pi^{FB}$.

Proof of Proposition D2. A constant threshold rule that selects a status quo action guarantees Π^{SQ} in every equilibrium. Hence

$$\Pi^{SQ} \leq \underline{\Pi}_n^T \leq \underline{\Pi}_n \leq \Pi(n) \leq \Pi^{FB}. \tag{D.3}$$

Statements 1 and 2, and the first claim in Statement 3, follow immediately from Proposition 3 and (D.3).

For the lower bound in Statement 3, let $t := F'(0) > 0$, $s := F(0) \in (0, 1)$, $\sigma^2 := s(1-s)$, and $\ell := |L|$. Consider the threshold

$$T_n := \lceil ns \rceil, \qquad d_n(k) := \mathbf{1}\{k \geq T_n\}.$$

Fix any symmetric equilibrium under d_n and write

$$p_{n,\omega} := \Pr(r_i = 1 \mid \omega), \qquad \rho_{n,\omega} := \Pr(\mathrm{Bin}(n-1, p_{n,\omega}) = T_n - 1)\,.$$

By Lemma 2, the gross benefit from reporting, which is $H\rho_{n,1}$ in state 1 and $L\rho_{n,0}$ in state 0, is at most $M/\sqrt{n}$ in absolute value. Since F is continuously differentiable around zero, it has no atoms there, so for all sufficiently large n no type is indifferent with positive probability; hence

$$p_{n,1} = F(H\rho_{n,1}), \qquad p_{n,0} = F(L\rho_{n,0}), \tag{D.4}$$

and, uniformly over equilibria,

$$p_{n,\omega} = s + O(n^{-1/2}).$$

Define

$$u_{n,\omega} := \frac{T_n - np_{n,\omega}}{\sigma\sqrt{n}}.$$

These variables are uniformly bounded. Lemma C3, applied as in the proof of Lemma A1,

gives, uniformly over equilibria,

$$\sqrt{n}\,\rho_{n,\omega} - \frac{1}{\sigma}\varphi(u_{n,\omega}) \longrightarrow 0, \qquad \Pr(\mathrm{Bin}(n, p_{n,\omega}) \geq T_n) - \Phi(-u_{n,\omega}) \longrightarrow 0.$$

Since $F(x) = s + tx + o(x)$ and $(T_n - ns)/(\sigma\sqrt{n}) \to 0$, (D.4) implies, uniformly over equilibria,

$$u_{n,1} + \frac{tH}{\sigma^2}\varphi(u_{n,1}) \longrightarrow 0, \qquad u_{n,0} - \frac{t\ell}{\sigma^2}\varphi(u_{n,0}) \longrightarrow 0.$$

For $x > 0$, let $z_x(t) > 0$ be the unique solution of

$$z = \frac{tx}{\sigma^2}\varphi(z).$$

The map $z \mapsto z/\varphi(z)$ is a strictly increasing bijection of $\mathbb{R}$ with a continuous inverse. Since the $u_{n,\omega}$ are bounded, dividing the preceding limits by $\varphi(u_{n,\omega})$ gives, uniformly over equilibria, $u_{n,1} \to -z_H(t)$ and $u_{n,0} \to z_\ell(t)$. Since

$$\mathbb{E}_{e,d_n}[u^p(a,\omega)] = \Pi^{SQ} + \pi H \Pr(a = 1 \mid \omega = 1) - (1-\pi)\ell \Pr(a = 1 \mid \omega = 0),$$

it follows that

$$\lim_{n\to\infty} \inf_{e \in \mathcal{E}(d_n)} \mathbb{E}_{e,d_n}[u^p(a,\omega)] = \Pi^{SQ} + J(t),$$

where

$$J(t) := \pi H \Phi(z_H(t)) - (1-\pi)\ell\Phi(-z_\ell(t)).$$

Together with the constant status quo rule, this yields

$$\liminf_{n\to\infty} \underline{\Pi}_n^T \geq \Pi^{SQ} + \max\{0, J(t)\}.$$

The function $z_x(t)$ is continuous and strictly increasing in t, tends to zero as $t \downarrow 0$, and diverges as $t \to \infty$. Hence J is continuous and strictly increasing, with

$$\lim_{t\downarrow 0} J(t) = \frac{1}{2}\big[\pi H - (1-\pi)\ell\big] < 0, \qquad \lim_{t\to\infty} J(t) = \pi H > 0.$$

There is therefore a unique $\underline{F}_0 \in (0,\infty)$ such that $J(\underline{F}_0) = 0$. It depends only on π, H, L, and $F(0)$. This proves the second claim in Statement 3.

For Statement 4, write

$$s_- := F(0^-), \qquad s_+ := F(0),$$

so that $0 < s_- < s_+ < 1$. Choose $\tau \in (s_-, s_+)$ and use the threshold $T_n := \lceil n\tau \rceil$. For all

sufficiently large n, $1 < T_n < n$. Lemma 1 then implies that the probability an individual is pivotal is nonzero in every state and every equilibrium. Therefore, every type with $\nu_i \leq 0$ strictly prefers to report in state 1, and every type with $\nu_i \geq 0$ strictly prefers not to report in state 0, so, uniformly over equilibria,

$$p_{n,1} \geq s_+ > \tau > s_- \geq p_{n,0}.$$

Let $\eta := \min\{s_+ - \tau, \tau - s_-\} > 0$. Since $T_n/n \to \tau$ and

$$\operatorname{Var}\left(\frac{1}{n}\sum_{i=1}^{n} r_i \,\middle|\, \omega\right) \leq \frac{1}{4n},$$

Chebyshev's inequality implies, uniformly over equilibria,

$$\Pr(a = 0 \mid \omega = 1) + \Pr(a = 1 \mid \omega = 0) \leq \frac{2}{n\eta^2}$$

for all sufficiently large n. Thus the worst-equilibrium payoff under this threshold converges to Π^{FB}. Equation (D.3) completes the proof. □

D.3 Noisy Observation of the State

Throughout, I maintain the general model in Section 3 and Assumption 1, except that individuals do not observe ω directly. Conditional on $\omega \in \Omega \subseteq \mathbb{R}$, individual i observes a signal $s_i \in \mathbb{R}$, where the signals are i.i.d. according to $G(\cdot \mid \omega)$ and are independent of the reporting costs. A symmetric reporting strategy is a function

$$(d, s_i, \nu_i) \mapsto \sigma_n(d, s_i, \nu_i) \in \Delta(\{0, 1\}).$$

Let $\mathcal{E}^G(d)$ denote the set of symmetric Bayesian Nash equilibria under this information structure and define

$$\Pi^G(n) := \sup_{d \in D(n),\, e \in \mathcal{E}^G(d)} \mathbb{E}_{e,d,G}[u^p(a, \omega)].$$

The first-best benchmark Π^{FB} remains the payoff available when the decision-maker observes ω directly.

Proposition D3 (Noisy Observation of the State)**.** *Under Assumption 1, the following statements hold for any signal structure $G(\cdot \mid \omega)$:*

1. *If $F(\delta) - F(-\delta) = 0$ for some $\delta > 0$, then there exists a finite n^* such that $\Pi^G(n) = \Pi^{SQ}$ for every $n > n^*$.*

2. *If F is differentiable at zero and $F'(0) = 0$, then $\Pi^G(n) \to \Pi^{SQ}$.*

3. *If F is differentiable at zero and $F'(0) > 0$, then $\limsup_{n\to\infty} \Pi^G(n) < \Pi^{FB}$.*

Proof of Proposition D3. Fix n, a symmetric decision rule, and a symmetric equilibrium. The argument proving Lemma 1 is unchanged: conditional on any signal, types with $\nu_i < -B$ strictly prefer to report, and types with $\nu_i > B$ strictly prefer not to report. Hence, for some $\epsilon_* > 0$,

$$p_{n,\omega} := \Pr(r_i = 1 \mid \omega) \in [\epsilon_*, 1 - \epsilon_*]$$

for every state, decision rule, and equilibrium. By symmetry and conditional independence, conditional on ω the reports are i.i.d. Bernoulli with parameter $p_{n,\omega}$.

Let $\Delta_{n,\omega}$ denote the state-contingent gross benefit from changing one report from 0 to 1. Following the proof of Lemma 2, there exists an $M > 0$, uniform in the state, decision rule, equilibrium, and signal structure, such that

$$|\Delta_{n,\omega}| \leq h_n := \frac{M}{\sqrt{n}}.$$

By conditional independence, after observing $s_i = s$ the gross benefit from reporting is

$$\Delta_n(s) := \mathbb{E}[\Delta_{n,\omega} \mid s_i = s].$$

Since $|\Delta_{n,\omega}| \leq h_n$ in every state, we have $|\Delta_n(s)| \leq h_n$. Since the net benefit from reporting is $\Delta_n(s) - \nu_i$, allowing a type at the cutoff to mix gives

$$F((-h_n)^-) \leq \Pr(r_i = 1 \mid s_i = s) \leq F(h_n)$$

for every s. Averaging over signals conditional on ω yields

$$F((-h_n)^-) \leq p_{n,\omega} \leq F(h_n). \tag{D.5}$$

This is the same reporting-probability bound used in the proof of Proposition 4. Since, conditional on ω, the report count is $\text{Bin}(n, p_{n,\omega})$, the proofs of Statements 1–3 of that proposition apply without change. □

The previous proposition concerns only the upper bounds on the decision-maker's payoff. The lower bound in Statement 4 of Proposition 4 need not extend, because imperfect private signals may themselves prevent the decision-maker from attaining the payoff available under direct observation of ω; see Barelli et al. (2022). As an extreme example, if s_i were

independent of ω, then reports would be independent of the state, so $\Pi^G(n) = \Pi^{SQ}$ for every n.

D.4 General Message Spaces

I consider the general environment of Section 3, rather than the binary state and action specialization in Section 4. For this subsection, I will replace Assumption 1 with the mechanical-type structure specified below, which generalizes the one in Section 2. Here, F denotes the CDF of ν_i conditional on the individual being rational. Define

$$B := \max_{a,\omega} u^p(a,\omega) - \min_{a,\omega} u^p(a,\omega).$$

Let $\mathcal{X} := \{\varnothing\} \cup [0,1]$, where $\varnothing$ denotes no message. Each individual is independently mechanical with probability $\epsilon \in (0,1)$ and rational with probability $1-\epsilon$. An individual's type is unobserved by the decision-maker. A mechanical individual sends $\varnothing$ with probability $\eta \in (0,1)$ and otherwise draws a message uniformly from $[0,1]$, independently of the state and all other uncertainty. A rational individual privately observes a cost $\nu_i \geq 0$, drawn independently according to F, and chooses a message $m_i \in \mathcal{X}$. His utility is

$$u^i(a,\omega,\nu_i,m_i) = u^p(a,\omega) - \nu_i \mathbf{1}_{m_i \neq \varnothing}. \tag{D.6}$$

The decision-maker observes $y_i \in \mathcal{X}$. If $m_i = \varnothing$, then $y_i = \varnothing$. If $m_i = m \in [0,1]$, then $y_i \in [0,1]$ has density $k(y \mid m)$. Conditional on m_i, the observation y_i is drawn independently of all other uncertainty. A decision rule is a measurable map $d_n : \mathcal{X}^n \to \Delta(A)$ that is invariant under permutations of its arguments, which is the generalization of the symmetry assumption.

Let $D^{\mathcal{M}}(n)$ denote the set of these decision rules, let $\mathcal{E}^{\mathcal{M}}(d_n)$ denote the set of symmetric Bayesian Nash equilibria among the rational individuals, with independent private randomization, and define

$$\Pi^{\mathcal{M}}(n) := \sup_{d_n \in D^{\mathcal{M}}(n),\ e \in \mathcal{E}^{\mathcal{M}}(d_n)} \mathbb{E}_{e,d_n}[u^p(a,\omega)]. \tag{D.7}$$

Assumption 2 (Noisy Recording)**.** *For every $m \in [0,1]$, $k(\cdot \mid m)$ is a density on $[0,1]$, and $k(y \mid m)$ is continuous and strictly positive on $[0,1]^2$.*

Assumption 2 allows recorded messages to be concentrated near the intended message but requires every recorded message to remain possible. For example, it is satisfied by a truncated-normal recording error centered at m. Assumption 2 and compactness imply that

there exist constants $0 < \underline{k} \leq \overline{k} < \infty$ such that

$$\underline{k} \leq k(y \mid m) \leq \overline{k} \qquad \text{for every } m, y \in [0,1]. \tag{D.8}$$

Remark 2 (Continuum Messages)**.** *With a continuum of messages, mechanical individuals alone are not sufficient. Without Assumption 2, the decision rule whereby the decision-maker chooses $a = 1$ if and only if she observes two identical nonnull messages effectively removes the noise and results in a non-noisy Volunteer's Dilemma with a threshold of two for action.*

The above remark relies on perfect coordination and thus, arguably, relies too heavily on common knowledge.

Remark 3 (Finite Messages)**.** *If $\mathcal{X}$ is finite and mechanical behavior assigns positive probability to every message, Assumption 2 is not needed. The proofs below use that assumption only to obtain a constant Λ such that, for every $m \in \mathcal{X}$, the distribution of the observed message when an individual sends m is at most $\Lambda\epsilon$ times the distribution of a mechanical individual's observed message. If the messages sent are the messages received (i.e., Assumption 2 fails), χ is the distribution of messages sent by mechanical individuals, and $\underline{\chi} := \min_{m \in \mathcal{X}} \chi(m) > 0$, this property holds with $\Lambda = (\epsilon\underline{\chi})^{-1}$, so the proofs below apply.*

Define

$$\Lambda := \max\left\{\frac{1}{\epsilon\eta}, \frac{\overline{k}}{\epsilon(1-\eta)\underline{k}}\right\}.$$

Lemma D1 (Anonymous Influence)**.** *Fix a state ω, a symmetric decision rule, and a symmetric strategy profile for the rational individuals. Let $U_{n,\omega}(m)$ denote a rational individual's expected value of $u^p(a,\omega)$ when he sends $m \in \mathcal{X}$, holding fixed the strategies of all other individuals. Then, for every $m, m' \in \mathcal{X}$,*

$$|U_{n,\omega}(m) - U_{n,\omega}(m')| \leq \frac{B\sqrt{\Lambda - 1}}{\sqrt{n}}. \tag{D.9}$$

The bound is uniform over states, symmetric decision rules, and symmetric strategy profiles.

Proof of Lemma D1. Fix ω, the decision rule, and the symmetric strategy profile. Conditional on ω, the observed messages generated by the other individuals are independent and have a common distribution P_ω. Write

$$q_\omega := P_\omega(\{\varnothing\})$$

and let p_ω denote the density of P_ω on $[0,1]$. Mechanical behavior implies

$$q_\omega \geq \epsilon\eta, \tag{D.10}$$

$$p_\omega(y) \geq \epsilon(1-\eta)\int_0^1 k(y \mid t)\,dt \geq \epsilon(1-\eta)\underline{k} \qquad \text{for every } y \in [0,1]. \tag{D.11}$$

For $m \in \mathcal{X}$, let Q_m denote the distribution of the observed message generated when an individual sends the intended message m. Thus, $Q_\varnothing$ is the point mass at $\varnothing$, while for $m \in [0,1]$, Q_m has density $k(\cdot \mid m)$ on $[0,1]$. Define the likelihood ratio $w_m = dQ_m/dP_\omega$ by

$$w_\varnothing(z) := \begin{cases} 1/q_\omega, & z = \varnothing, \\ 0, & z \in [0,1], \end{cases}$$

$$w_m(z) := \begin{cases} 0, & z = \varnothing, \\ k(z \mid m)/p_\omega(z), & z \in [0,1], \end{cases} \qquad m \in [0,1].$$

Equation (D.11) and the upper bound $k(y \mid m) \leq \overline{k}$ imply

$$0 \leq w_m \leq \Lambda \qquad \text{for every } m \in \mathcal{X}.$$

Moreover, if $Y \sim P_\omega$, then

$$\mathbb{E}[w_m(Y)] = 1 \qquad \text{for every } m \in \mathcal{X}. \tag{D.12}$$

Let $Y_1, \ldots, Y_n$ be independent random variables with common distribution P_ω, and define

$$V_\omega(Y_1, \ldots, Y_n) := \int_A u^p(a,\omega)\, d_n(Y_1, \ldots, Y_n)(da),$$
$$W_m := \frac{1}{n}\sum_{j=1}^n w_m(Y_j).$$

For every j,

$$\mathbb{E}[V_\omega(Y_1, \ldots, Y_n) w_m(Y_j)] = \int V_\omega(y^n)\, Q_m(dy_j) \prod_{\ell \neq j} P_\omega(dy_\ell).$$

The right-hand side is the expected gross payoff obtained when one individual sends m and the other $n-1$ observations have distribution P_ω. By symmetry of the decision rule, it is independent of j and equals $U_{n,\omega}(m)$. Averaging over j therefore gives

$$U_{n,\omega}(m) = \mathbb{E}[V_\omega W_m]. \tag{D.13}$$

Equation (D.12) and independence imply $\mathbb{E}[W_m] = 1$ and

$$\operatorname{Var}(W_m) = \frac{1}{n}\operatorname{Var}(w_m(Y)) = \frac{1}{n}\left(\mathbb{E}[w_m(Y)^2] - 1\right) \leq \frac{\Lambda - 1}{n},$$

where the inequality follows from $w_m(Y)^2 \leq \Lambda w_m(Y)$.

Let

$$\overline{U}_{n,\omega} := \mathbb{E}[V_\omega].$$

The range of V_ω has length at most B, and hence $\operatorname{Var}(V_\omega) \leq B^2/4$. Using Equation (D.13),

$$\left|U_{n,\omega}(m) - \overline{U}_{n,\omega}\right| = |\operatorname{Cov}(V_\omega, W_m)| \leq \sqrt{\operatorname{Var}(V_\omega)\operatorname{Var}(W_m)} \leq \frac{B\sqrt{\Lambda - 1}}{2\sqrt{n}}.$$

Applying this inequality to m and m' and using the triangle inequality yields

$$|U_{n,\omega}(m) - U_{n,\omega}(m')| \leq \frac{B\sqrt{\Lambda - 1}}{\sqrt{n}}.$$

Since B and Λ depend only on the primitives, this bound is uniform over states, symmetric decision rules, and symmetric strategy profiles. □

Let $M_G := B\sqrt{\Lambda - 1}$ and define

$$q_{n,\omega} := \Pr(m_i \neq \varnothing \mid \omega,\ i \text{ is rational}).$$

Because the null message is free, every rational individual with $\nu_i > M_G/\sqrt{n}$ strictly prefers $\varnothing$ to every nonnull message. Consequently, in every state and every symmetric equilibrium,

$$q_{n,\omega} \leq F\left(\frac{M_G}{\sqrt{n}}\right). \tag{D.14}$$

Proposition D4 (General Message Spaces)**.** *Under Assumption 2, the following hold:*

1. *If there exists $\delta > 0$ such that $F(\delta) = 0$, then there exists $n^* \in \mathbb{N}$ such that $\Pi^{\mathcal{M}}(n) = \Pi^{SQ}$ for every $n > n^*$.*

2. *Suppose $F(0) = 0$ and F is right-differentiable at zero. If $F'_+(0) = 0$, then*

$$\lim_{n\to\infty} \Pi^{\mathcal{M}}(n) = \Pi^{SQ}.$$

3. Suppose $F(0) = 0$ and $0 < F'_+(0) < \infty$. Then

$$\limsup_{n\to\infty} \Pi^{\mathcal{M}}(n) < \Pi^{FB}.$$

Proof of Proposition D4. Equation (D.14) gives $q_{n,\omega} \leq F(M_G/\sqrt{n})$ uniformly over states, symmetric decision rules, and symmetric equilibria.

For Statement 1, choose n^* such that $M_G/\sqrt{n} < \delta$ for every $n > n^*$. Then $q_{n,\omega} = 0$ in every state and every symmetric equilibrium. The observation profile is therefore independent of the state, so every decision rule yields at most Π^{SQ}. A constant rule attains Π^{SQ}, proving the claim.

For Statements 2 and 3, fix n, $d_n \in D^{\mathcal{M}}(n)$, and $e \in \mathcal{E}^{\mathcal{M}}(d_n)$. Let P^0 be the distribution of an individual's observed message when every rational individual sends $\varnothing$, which does not depend on the state.

Let P_ω and Q_m be as in the proof of Lemma D1. In state ω, a rational individual's observed message has distribution $(1 - q_{n,\omega})Q_\varnothing + q_{n,\omega}S_{n,\omega}$, where $S_{n,\omega}$ is a mixture of the distributions Q_m, $m \in [0, 1]$. Hence

$$P_\omega - P^0 = (1 - \epsilon)q_{n,\omega}\big(S_{n,\omega} - Q_\varnothing\big).$$

Since P^0 is the distribution P_ω under the strategy profile in which every rational individual sends $\varnothing$, the bound $w_m \leq \Lambda$ from that proof gives $Q_m \leq \Lambda P^0$ for every $m \in \mathcal{X}$, and hence $S_{n,\omega} \leq \Lambda P^0$. Therefore, relative to P^0, P_ω has a likelihood ratio $r_{n,\omega}$ such that, with $C_0 := (1 - \epsilon)\Lambda$,

$$|r_{n,\omega}(z) - 1| \leq C_0 q_{n,\omega} \qquad \text{for every } z \in \mathcal{X}. \tag{D.15}$$

Conditional on ω, the observed messages $y_1, \ldots, y_n$ are independent and identically distributed, since rational individuals use a common strategy and individuals' types, costs, randomizations, and recording errors are independent. Let $\mathbb{P}_{n,\omega}$ denote their joint distribution and let $\mathbb{P}^0_n$ denote the joint distribution of n independent draws from P^0. The likelihood ratio of $\mathbb{P}_{n,\omega}$ relative to $\mathbb{P}^0_n$ is

$$L_{n,\omega}(y^n) := \prod_{i=1}^{n} r_{n,\omega}(y_i).$$

Define

$$Z_\omega(y^n) := \max_{a\in A} u^p(a, \omega) - \int_A u^p(a, \omega)\, d_n(y^n)(da), \qquad 0 \leq Z_\omega \leq B,$$

and let

$$R_{n,\omega} := \mathbb{E}_{\mathbb{P}_{n,\omega}}[Z_\omega], \qquad R^0_{n,\omega} := \mathbb{E}_{\mathbb{P}^0_n}[Z_\omega].$$

Since $\mathbb{P}^0_n$ is independent of the state, the action distribution induced by d_n under $\mathbb{P}^0_n$ yields at most Π^{SQ}. Hence, with $\Delta := \Pi^{FB} - \Pi^{SQ} > 0$,

$$\mathbb{E}_\omega[R^0_{n,\omega}] \geq \Delta.$$

For Statement 2, $F(0) = 0$ and a zero right derivative imply $\sqrt{n}\, F(M_G/\sqrt{n}) \to 0$, so $nq^2_{n,\omega} \to 0$ uniformly. Because $r_{n,\omega}$ is a likelihood ratio, $\mathbb{E}_{P^0}[r_{n,\omega}] = 1$. Independence therefore gives

$$\mathbb{E}_{\mathbb{P}^0_n}[(L_{n,\omega} - 1)^2] = \left(1 + \mathbb{E}_{P^0}[(r_{n,\omega} - 1)^2]\right)^n - 1 \leq e^{nC_0^2 q^2_{n,\omega}} - 1 \longrightarrow 0$$

uniformly. Consequently,

$$|R_{n,\omega} - R^0_{n,\omega}| \leq B\left(\mathbb{E}_{\mathbb{P}^0_n}[(L_{n,\omega} - 1)^2]\right)^{1/2} = o(1)$$

uniformly, and hence

$$\mathbb{E}_{e,d_n}[u^p(a,\omega)] = \Pi^{FB} - \mathbb{E}_\omega[R_{n,\omega}] \leq \Pi^{SQ} + o(1).$$

Taking the supremum over d_n and e, and recalling that a constant rule can achieve Π^{SQ}, proves Statement 2.

For Statement 3, Equation (D.14), $F(0) = 0$, and a finite right derivative imply that there is $C_1 < \infty$, independent of ω, d_n, and e, such that $q_{n,\omega} \leq C_1/\sqrt{n}$ for all sufficiently large n. By Equation (D.15), $|r_{n,\omega} - 1| \leq C/\sqrt{n}$ with $C := C_0 C_1$, and $C/\sqrt{n} \leq 1/2$ for all sufficiently large n.

Since $(1+x)^{-1} \leq 1 - x + 2x^2$ for $|x| \leq 1/2$ and $\mathbb{E}_{P^0}[r_{n,\omega} - 1] = 0$,

$$\mathbb{E}_{\mathbb{P}^0_n}\left[\frac{1}{L_{n,\omega}}\right] = \left(\mathbb{E}_{P^0}\left[\frac{1}{r_{n,\omega}}\right]\right)^n \leq \left(1 + \frac{2C^2}{n}\right)^n \leq e^{2C^2}.$$

Since $L_{n,\omega}$ is the likelihood ratio of $\mathbb{P}_{n,\omega}$ relative to $\mathbb{P}^0_n$, we have $R_{n,\omega} = \mathbb{E}_{\mathbb{P}^0_n}[Z_\omega L_{n,\omega}]$, and $L_{n,\omega} > 0$ because $r_{n,\omega} \geq 1/2$. Cauchy–Schwarz under $\mathbb{P}^0_n$ therefore gives

$$(R^0_{n,\omega})^2 = \left(\mathbb{E}_{\mathbb{P}^0_n}\left[\sqrt{Z_\omega L_{n,\omega}}\sqrt{\frac{Z_\omega}{L_{n,\omega}}}\right]\right)^2 \leq R_{n,\omega}\, \mathbb{E}_{\mathbb{P}^0_n}\left[\frac{Z_\omega}{L_{n,\omega}}\right] \leq Be^{2C^2} R_{n,\omega}.$$

Averaging over states and applying Jensen's inequality,

$$\Pi^{FB} - \mathbb{E}_{e,d_n}[u^p(a,\omega)] \geq \frac{e^{-2C^2}}{B}\mathbb{E}_{\omega}[(R^0_{n,\omega})^2] \geq \frac{e^{-2C^2}}{B}\Delta^2 > 0.$$

The bound is uniform over d_n and e, so

$$\limsup_{n\to\infty} \Pi^{\mathcal{M}}(n) < \Pi^{FB}. \qquad \square$$